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Universal Design of ICT

Designing Playful and Accessible Learning  
Experiences to Enhance AI Literacy for Older  
Adults

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## PREFACE

Working on such an important topic for my thesis has been incredibly rewarding. To explore solutions that might prevent people from getting left behind in an increasingly digitalized world proved both a privilege and a challenge I am grateful to have taken on.

I want to thank all the participants who took part in this study. Their willingness to try something new, and possibly a little uncomfortable, has made this project possible. A special thanks goes to Seniornett Sarpsborg Øst for welcoming me into their datacafé with such openness from the very first week.

I would also like to thank my supervisor, Weiqin Chen, for her guidance and support throughout this process.

Finally, thank you to my family, and especially to my dear Veslemøy, for your endless love and encouragement. And to my cat, Nyx, for purring reassuringly by my side when things got rough.

Sarpsborg, 14.08.2026

Alexander Mora

## ABSTRACT

Older adults face growing risk of digital exclusion as AI becomes part of everyday products and services, and the aging population of Norway means this risk will affect a larger share of society in the coming decades. Existing AI literacy resources do not fit this group well, as most target younger or general audiences, cover only a single AI product, or leave out AI-related harms like scams and misinformation.

This thesis addresses that gap with SeniorKI, a self-paced web platform teaching AI literacy to Norwegian adults aged 60 and above, built around the five modules of the curriculum proposed by Kaur et al. (2025), with some gamification elements. The platform was developed using an iterative, user-centered design process and evaluated through two rounds of formative usability testing and a summative evaluation with seven completers, combining a knowledge test, the System Usability Scale (SUS), and qualitative feedback.

Results were mixed. Five of seven participants improved their knowledge test scores, a statistically significant result under a one-sided test ( $p = .031$ ), though this did not hold two-sided or when a participant with a possible information transfer concern was excluded. The platform scored highly on usability (SUS = 89.3, within the “Best imaginable” on standard SUS benchmarks), and formative usability testing resolved several usability problems before the final evaluation. The summative evaluation lasted for two weeks, but no participant used the platform for more than two days. This meant the evaluation could not show whether its gamification elements actually encourage continued use, which was their intended purpose.

Three of the project’s four objectives were substantially met, while the fourth, on gamification and engagement, was explored but left largely unanswered. The platform also deviates from its source curriculum in two ways, both due to time constraints: it does not include the hands-on activity specified for each module, and it is only available online, rather than as a blended learning format that the curriculum proposes. Future work should prioritize adding both elements, since both this project’s findings and existing literature both

suggest hands-on and in-person support may matter most for the least digitally confident users.

Aside from its research contributions, SeniorKI is a working, public platform rather than a prototype. The findings and design choices documented here, especially around usability, may be useful for future AI literacy tools built for older adults, a group that is underrepresented in AI literacy research.

# TABLE OF CONTENTS

|                                                                        |           |
|------------------------------------------------------------------------|-----------|
| <b>PREFACE .....</b>                                                   | <b>2</b>  |
| <b>ABSTRACT .....</b>                                                  | <b>3</b>  |
| <b>TABLE OF CONTENTS .....</b>                                         | <b>5</b>  |
| <b>LIST OF FIGURES .....</b>                                           | <b>8</b>  |
| <b>1 INTRODUCTION .....</b>                                            | <b>9</b>  |
| 1.1 Research aim and research questions .....                          | 10        |
| 1.2 Project scope and limitations .....                                | 12        |
| 1.3 Ethical considerations.....                                        | 13        |
| 1.4 Thesis structure.....                                              | 14        |
| <b>2 LITERATURE REVIEW .....</b>                                       | <b>16</b> |
| 2.1 Prior study.....                                                   | 16        |
| 2.2 AI literacy.....                                                   | 17        |
| 2.3 Universal design of ICT and interface design for older adults..... | 18        |
| 2.3.1 Universal design of ICT .....                                    | 18        |
| 2.3.2 Interface design for older adults .....                          | 19        |
| 2.4 Gamification in learning platforms .....                           | 20        |
| <b>3 METHODOLOGY.....</b>                                              | <b>22</b> |
| 3.1 Research approach .....                                            | 22        |
| 3.2 Development process .....                                          | 23        |
| Paper prototyping .....                                                | 23        |
| Figma design .....                                                     | 24        |
| React implementation.....                                              | 24        |
| Curriculum content development.....                                    | 25        |
| 3.3 Evaluation.....                                                    | 25        |
| 3.3.1 Recruitment.....                                                 | 25        |
| 3.3.2 Formative evaluation.....                                        | 26        |
| 3.3.3 Summative evaluation .....                                       | 28        |
| 3.4 Data analysis .....                                                | 30        |
| 3.4.1 Qualitative data analysis.....                                   | 30        |
| 3.4.2 Quantitative data analysis .....                                 | 31        |
| 3.5 Ethics and data management .....                                   | 31        |
| <b>4 RESULTS .....</b>                                                 | <b>32</b> |
| 4.1 Platform overview .....                                            | 32        |
| 4.1.1 Platform presentation .....                                      | 33        |
| 4.2 Iterative design and development.....                              | 46        |

|          |                                                                                                                                                                                                                                                                                         |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4.2.1    | Iteration 1: Figma prototype .....                                                                                                                                                                                                                                                      | 46        |
| 4.2.2    | Iteration 2: Coded prototype.....                                                                                                                                                                                                                                                       | 50        |
| 4.2.3    | Design changes summary table .....                                                                                                                                                                                                                                                      | 53        |
| 4.3      | <i>Final platform design</i> .....                                                                                                                                                                                                                                                      | 55        |
| 4.3.1    | Accessibility.....                                                                                                                                                                                                                                                                      | 55        |
| 4.3.2    | Visual design .....                                                                                                                                                                                                                                                                     | 58        |
| 4.3.3    | Gamification elements .....                                                                                                                                                                                                                                                             | 61        |
|          | Stars.....                                                                                                                                                                                                                                                                              | 61        |
|          | Badges.....                                                                                                                                                                                                                                                                             | 61        |
|          | Progress overview .....                                                                                                                                                                                                                                                                 | 61        |
| 4.4      | <i>Curriculum structure</i> .....                                                                                                                                                                                                                                                       | 62        |
|          | Module 1: Introduction to AI (Introduksjon til KI) .....                                                                                                                                                                                                                                | 63        |
|          | Module 2: Building confidence with AI (Bygge selvtillitt med KI) .....                                                                                                                                                                                                                  | 63        |
|          | Module 3: Generative AI (Generativ KI).....                                                                                                                                                                                                                                             | 63        |
|          | Module 4: AI in smart homes (KI i smarthjem) .....                                                                                                                                                                                                                                      | 63        |
|          | Module 5: Staying safe with AI (Hold deg trygg med KI) .....                                                                                                                                                                                                                            | 63        |
| 4.5      | <i>Summative evaluation</i> .....                                                                                                                                                                                                                                                       | 64        |
| 4.5.1    | Quantitative evaluation results .....                                                                                                                                                                                                                                                   | 64        |
| 4.5.2    | Qualitative evaluation results.....                                                                                                                                                                                                                                                     | 65        |
| <b>5</b> | <b>DISCUSSION.....</b>                                                                                                                                                                                                                                                                  | <b>67</b> |
| 5.1      | <i>Results discussion</i> .....                                                                                                                                                                                                                                                         | 67        |
| 5.1.1    | Demographic information .....                                                                                                                                                                                                                                                           | 67        |
| 5.1.2    | Action paralysis among the least experienced participants.....                                                                                                                                                                                                                          | 68        |
| 5.1.3    | P7/P13 knowledge gain .....                                                                                                                                                                                                                                                             | 69        |
| 5.1.4    | Completion times and self-paced use .....                                                                                                                                                                                                                                               | 70        |
| 5.1.5    | SUS responses.....                                                                                                                                                                                                                                                                      | 71        |
| 5.1.6    | Gamification elements .....                                                                                                                                                                                                                                                             | 71        |
| 5.1.7    | Reflections on AI-assisted development .....                                                                                                                                                                                                                                            | 72        |
| 5.2      | <i>Limitations</i> .....                                                                                                                                                                                                                                                                | 73        |
| 5.2.1    | Bias.....                                                                                                                                                                                                                                                                               | 73        |
| 5.2.2    | Small sample sizes .....                                                                                                                                                                                                                                                                | 74        |
| 5.2.3    | Knowledge test ceiling and answer distribution .....                                                                                                                                                                                                                                    | 75        |
| 5.2.4    | Attrition .....                                                                                                                                                                                                                                                                         | 75        |
| 5.2.5    | Single researcher qualitative analysis .....                                                                                                                                                                                                                                            | 76        |
| 5.2.6    | Effect of the test situation on observed behavior.....                                                                                                                                                                                                                                  | 76        |
| 5.2.7    | Small font size .....                                                                                                                                                                                                                                                                   | 77        |
| 5.2.8    | Accessibility testing methodology .....                                                                                                                                                                                                                                                 | 77        |
| 5.3      | <i>Future work</i> .....                                                                                                                                                                                                                                                                | 78        |
| 5.3.1    | Interactive creative activities .....                                                                                                                                                                                                                                                   | 78        |
| 5.3.2    | Blended learning.....                                                                                                                                                                                                                                                                   | 78        |
| 5.3.3    | Social sharing of results .....                                                                                                                                                                                                                                                         | 79        |
| <b>6</b> | <b>CONCLUSION .....</b>                                                                                                                                                                                                                                                                 | <b>80</b> |
|          | <b>REFERENCE LIST.....</b>                                                                                                                                                                                                                                                              | <b>83</b> |
|          | <b>APPENDICES.....</b>                                                                                                                                                                                                                                                                  | <b>88</b> |
| 1.       | <i>Appendix A: SIKT approval Confirmation of ethics approval (reference number: 697349) granted by Sikt (the Norwegian Agency for Shared Services in Education and Research) prior to data collection for the summative evaluation. See attached file/image: SIKT_approval.pdf.....</i> | <i>88</i> |

|            |                                                                                                                                                                                                                                                                                                                        |    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>2.</b>  | <i>Appendix B: Source code SeniorKI source code repository, tagged at the version corresponding to this thesis. Link: <a href="https://github.com/alexandermora1/ai-literacy-for-older-adults">https://github.com/alexandermora1/ai-literacy-for-older-adults</a>, commit f31b012, tag v1.0-thesis88</i>               |    |
| <b>3.</b>  | <i>Appendix C: Usability test protocol: Iteration 1 Task list and observation guide used during the first round of formative usability testing (Figma prototype), conducted with five participants. See attached file: Usability_test_protocol_1.pdf</i>                                                               | 88 |
| <b>4.</b>  | <i>Appendix D: Usability test protocol: Iteration 2 Task list and observation guide used during the second round of formative usability testing (coded prototype), conducted with three participants. See attached file: Usability_test_protocol_2.pdf</i>                                                             | 88 |
| <b>5.</b>  | <i>Appendix E: Pre-test questionnaire (Nettskjema) Blank pre-test questionnaire used in the summative evaluation, including the knowledge test and project information/consent text. See attached file: Nettskjema_pre.pdf</i>                                                                                         | 88 |
| <b>6.</b>  | <i>Appendix F: Post-test questionnaire (Nettskjema) Blank post-test questionnaire used in the summative evaluation, including the knowledge test, System Usability Scale, and open-ended feedback questions. See attached file: Nettskjema_post.pdf</i>                                                                | 88 |
| <b>7.</b>  | <i>Appendix G: Curriculum content: Original version Written curriculum content for all five modules, prior to language adaptation for the target group. See attached file: Curriculum_original.pdf</i>                                                                                                                 | 88 |
| <b>8.</b>  | <i>Appendix H: Curriculum content: Adapted version Written curriculum content for all five modules, following language adaptation for the target group. See attached file: Curriculum_adapted.pdf</i>                                                                                                                  | 89 |
| <b>9.</b>  | <i>Appendix I: Summative evaluation data Anonymized participant data from the summative evaluation, pseudonymized as P1–P9, including demographic information, pre/post knowledge test scores, System Usability Scale scores, and open-ended feedback responses. See attached file: Summative_evaluation_data.xlsx</i> | 89 |
| <b>10.</b> | <i>Appendix J: Language review prompt Prompt given to Claude (Anthropic) for the language-level review of curriculum content, describing the target group and intended tone, used as a fallback after the planned review by Seniornett course leaders did not take place due to time constraints.</i>                  | 89 |

## LIST OF FIGURES

|                                                                                                                                                        |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Figure 4.1: The seniorki.no welcome screen, shown on an Android tablet in landscape orientation.....                                                   | 33 |
| Figure 4.2: The onboarding/help page, upper portion, showing the course structure explanation and content sourcing.....                                | 34 |
| Figure 4.3: The onboarding/help page, lower portion, showing the text-size control and the button to begin or continue the course.....                 | 35 |
| Figure 4.4: The course overview page, showing six modules, with two of them completed, and a different number of stars earned.....                     | 36 |
| Figure 4.5: The topic overview page for the "Generativ KI" chapter, showing three numbered topic cards and a separated quiz card.....                  | 37 |
| Figure 4.6: A topic content page, showing the topic title, position within the chapter, estimated reading time, and body text.....                     | 39 |
| Figure 4.7: The same topic page, scrolled to the bottom, showing the "next topic" button has become available and the scroll bar has been filled. .... | 39 |
| Figure 4.8: Quiz page with an answer option selected, but not confirmed. ....                                                                          | 40 |
| Figure 4.9: Quiz feedback showing a correct answer, with the selected card turned green and labeled "Riktig!" .....                                    | 41 |
| Figure 4.10: Quiz feedback showing an incorrect answer, with both the user's selection and the correct answer highlighted. ....                        | 42 |
| Figure 4.11: The post quiz results page, upper portion, showing the star rating and newly earned badges. ....                                          | 43 |
| Figure 4.12: The post quiz results page, lower portion, showing the per question answer review.....                                                    | 43 |
| Figure 4.13: The badge earned page, showing a newly awarded badge.....                                                                                 | 44 |
| Figure 4.14: The progress page, upper portion, showing per-chapter progress bars. ....                                                                 | 45 |
| Figure 4.15: The progress overview page, lower portion, showing star and badge totals, and the badge collection. ....                                  | 46 |
| Figure 4.16: The Iteration 1 topic overview layout. Only two cards would be visible without scrolling on a tablet in landscape orientation.....        | 47 |
| Figure 4.17: The Iteration 1 module overview layout, showing the progress bar on each topic's card. ....                                               | 48 |

## LIST OF TABLES

|                                                                                                                                                               |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Table 1.1: Mapping of research questions to instruments, results sections, and discussion sections.....                                                       | 15 |
| Table 4.1: Summary of design changes made across iterations 1 and 2, showing the finding that prompted each change and where it was subsequently tested. .... | 54 |
| Table 4.2: Automated accessibility results.....                                                                                                               | 57 |
| Table 4.3: List of colors used in the platform.....                                                                                                           | 59 |



# 1 INTRODUCTION

The world's population is aging. The World Health Organization projects that by 2030, one in six people globally will be aged 60 or older, and that the number of people aged 60 and older worldwide will double to 2.1 billion by 2050 (WHO, 2025). This trend is similar in Norway. SSB projects that by the early 2030's, the number of people aged 65 and older will exceed the number of children and young people under 20 for the first time (Tømmerås et al., 2026). By the 2050's, the number of people who are 80 and older will more than double, and the number that are 90 and older will almost triple (Tømmerås et al., 2026).

As more people live into old age, artificial intelligence (AI) has become an increasingly prominent feature of modern life. From chatbots, to personalized recommendations on streaming services, to voice assistants in our homes, AI is now embedded in everyday products in ways most people will not notice. While this brings clear opportunities, it also raises the stakes for digital exclusion, a challenge to which older adults are considered more exposed than other groups in society (Sletteameås & Mainsah, 2025), and one that AI is likely to make worse. Unlike a website or an app, AI systems often make decisions invisibly, adapt their behavior without warning, and can be difficult to distinguish from a human on the other end. Older adults already report unease at not being sure whether they are interacting with a person or a machine, a concern that grows more common with age (Sletteameås, 2025). Many older adults lack the basic conceptual understanding to navigate AI-enabled products and services safely and with confidence, placing them at risk of falling further behind in a society that increasingly assumes digital participation.

This concern is reflected in the numbers: in a survey of 182 Norwegians aged 60 and above, Kaur and Chen (2023) found that while only 9 respondents (4.9%) reported knowing nothing about AI, a much larger group had only a shallow grasp of it, with 76 respondents saying they understood how AI works only to a small extent, compared to 33 who said they understood it to a large extent. This points to a gap between general awareness and

functional understanding, even as AI-enabled products and services become increasingly embedded in daily life. The risk this creates goes beyond a simple lack of access. Without sufficient knowledge of how AI works, older individuals may struggle to access and benefit from AI-enabled products and services, and this knowledge gap can itself increase distrust, anxiety, and vulnerability to misinformation, further widening the digital divide (Kaur et al., 2025). Taken together, this suggests that the more pressing challenge for older adults is not access to AI-enabled products, but the functional understanding needed to use them safely. This is the gap this thesis sets out to address by building AI literacy, understood here as the knowledge and critical evaluation skills needed to understand and engage safely with AI in daily life (Long & Magerko, 2020).

The artifact developed for this thesis is seniorki.no, a web-based AI literacy learning platform targeting Norwegian adults aged 60 and above. The platform makes the Kaur et al (2025) curriculum available in an accessible, self-paced online format, which incorporates gamification elements to encourage engagement and further use. It was developed using an iterative user-centered design process, involving usability feedback from members of the target group throughout the process.

## 1.1 Research aim and research questions

The main aim of this thesis is to design, develop, and evaluate an accessible AI literacy learning platform, based on the curriculum proposed by Kaur et al (2025), with gamification elements, tailored to Norwegian adults aged 60 and above. This included four related objectives:

1. To make an existing, research based AI literacy curriculum for older adults into a functional interactive platform.
2. To design the platform, using universal design principles and evidence-based guidelines for interface design, specifically for older adults.
3. To evaluate the platform's usability through iterative formative testing and a summative evaluation.

4. To explore whether the platform's gamification elements can encourage user engagement and continued use.

These aims are addressed through the following research question:

*To what extent does a self-paced online AI learning platform improve older adults' practical AI understanding, how usable do older adults find the platform, and to what extent do the platform's gamification elements encourage engagement and continued use?*

With the following sub-questions:

1. *What changes do participants show in practical AI understanding after using the platform?*
2. *How do participants rate the platform's usability, and which platform elements do they perceive as the most helpful or unhelpful?*
3. *What usability barriers are encountered, and what design changes would address them?*
4. *To what extent do participants engage with the platform over the evaluation period, and how do they respond to its gamification elements?*

This thesis makes the following contributions:

1. A five module Norwegian-language AI literacy curriculum adapted for self-paced online learning, based on Kaur et al. (2025).
2. A WCAG (Web Content Accessibility Guidelines) 2.1 AA compliant, responsive React web-app implementing the curriculum with gamification elements to encourage extended usage.
3. Empirical findings on changes in AI knowledge among Norwegian adults aged 60+, and their usability experience with the platform, measured through pre/post testing, System Usability Scale, and qualitative feedback.

## 1.2 Project scope and limitations

This project encompasses the design, development, and iterative usability evaluation of a web-based AI literacy course platform for Norwegian adults aged 60 or above. The platform implements the five-module curriculum proposed by Kaur et al. (2025), adapted for a self-paced online course format. The design process follows universal design principles with a particular focus on WCAG 2.1 AA compliance, and the platform was evaluated through three rounds of usability testing with members of the target group.

Several aspects were deliberately placed outside the scope of this project. The platform is not intended as a comprehensive AI education for all age groups, but specifically addresses the AI literacy needs of Norwegian older adults as defined by the curriculum in Kaur et al. (2025). It does not cover advanced or technical AI topics such as deep reinforcement learning or programming. The platform was not built to scale to enterprise or commercial deployment, and analytics, telemetry, or usage-tracking beyond simple client-side progress storage were intentionally excluded to minimize data collection and privacy risk. Long-term retention and behavior change beyond the two-week window of the summative evaluation were not measured, and the platform does not include the blended or in-person learning component that Kaur et al.'s (2025) curriculum proposes. Within the scope of this project, it is intended purely as a self-paced digital course. Finally, the platform's content and testing were conducted exclusively with a Norwegian-speaking population. Translation for other languages or cross-cultural validation of the curriculum were not undertaken.

The target group for this project is Norwegian adults aged 60 and above with a basic level of digital literacy, meaning individuals that have basic competence using a computer, tablet, or smartphone and accessing the internet, but who have limited prior knowledge of artificial intelligence. Participants were recruited through Seniornett Norge and through personal networks. The platform is not designed for individuals with severe cognitive or motor impairments that would prevent independent device use, and such individuals were not recruited for usability testing.

### 1.3 Ethical considerations

Research involving human participants requires careful ethical consideration, especially when working with older adults who may be more vulnerable to discomfort or confusion during testing sessions than younger participants, and for whom unfamiliarity with the technology being tested could be mistaken, by the participant, for personal failure. To protect participants' identities while allowing their contributions to be tracked across testing rounds, participants were pseudonymized as P1-P14, with numbering continuous across all three testing rounds.

Across both formative usability testing iterations (see section 4.2), participants were recruited on a voluntary basis and briefed verbally that it was the platform being tested, not their own ability, and that they could stop at any time without giving a reason. Despite this, some degree of discomfort during testing could not be fully avoided. For several participants, using an unfamiliar type of technology while being observed was itself an uncomfortable experience, independent of how the testing environment was managed. Being watched while attempting something they had low confidence in was, for some, simply an uncomfortable situation to be in. This was visible to varying degrees across sessions, despite efforts to keep the setting relaxed and low pressure. This is noted here as an inherent challenge of usability testing with participants that may have limited prior exposure to this kind of evaluation, rather than as a weakness in how the sessions themselves were conducted. No audio or video recordings were made, only written observation notes were taken, and no directly identifying information about participants was retained beyond what was needed to schedule the session, so no formal approval from SIKT (the Norwegian Agency for Shared Services in Education and Research) was needed.

Many participants, across all evaluations, were recruited through family connections or through repeated contact built up during volunteer work at Seniornett datacafés (see section 3.3.1). This raises an ethical consideration, where a participant with an existing personal or family tie to me may feel a degree of social pressure to either take part, to continue once started, or to soften negative feedback, that a participant with no such relation would not. This was addressed the same way as for all participants, by briefing them

that their participation was voluntary, that the platform rather than their own ability was being tested, and that they could withdraw at any time without consequence or needing to give a reason.

The summative evaluation (see section 4.5) involved a higher degree of data collection, since participants' pre- and post-test responses needed to be linked over a two-week period. This required temporarily storing participants names and email addresses, and the study was therefore submitted to SIKT prior to data collection, and approval was granted (see Appendix A). Data was collected using Nettskjema, the University of Oslo's GDPR-compliant survey tool, and stored on OsloMet's OneDrive in accordance with institutional data handling requirements. Participants were informed of the purpose of the study and their right to withdraw, and provided consent before beginning the pre-test. The key linking names to pseudonyms was stored separately from the response data and will be deleted once the project is completed.

## 1.4 Thesis structure

This thesis is structured as follows. Chapter 2 reviews the literature underlying the project covering Kaur et al.'s (2025) prior study, AI literacy, universal design and interface design for older adults, and gamification. Chapter 3 describes the methodology, covering the design and development process and the formative and summative evaluations. Chapter 4 presents the SeniorKI platform and the results of its development and evaluation. Chapter 5 discusses these results in relation to the project's research questions and existing literature, covers the project's limitations, and outlines directions for future work. Chapter 6 concludes the thesis by answering the research questions directly and summarizing the project's contributions.

**Main RQ:** To what extent does a self-paced online AI learning platform improve older adults' practical AI understanding, how usable do older adults find the platform, and to what extent do the platform's gamification elements encourage engagement and continued use?

| Sub-question                                                                                                                                    | Instrument / method                                                 | Results §       | Discussion §    |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------|-----------------|
| RQ1 – What changes do participants show in practical AI understanding after using the platform?                                                 | Pre/post knowledge test                                             | 4.5.1           | 5.1.3,<br>5.1.4 |
| RQ2 – How do participants rate the platform's usability, and which platform elements do they perceive as the most helpful or unhelpful?         | System Usability Scale, qualitative open-ended feedback             | 4.5.1,<br>4.5.2 | 5.1.5,<br>5.1.6 |
| RQ3 – What usability barriers are encountered, and what design changes would address them?                                                      | Usability testing (Iterations 1 and 2)                              | 4.2             | 5.1.2           |
| RQ4 – To what extent do participants engage with the platform over the evaluation period, and how to they respond to its gamification elements? | Completion time data, qualitative feedback on gamification elements | 4.5.1,<br>4.5.2 | 5.1.4,<br>5.1.6 |

Table 1.1: Mapping of research questions to instruments, results sections, and discussion sections.

The study involved 14 participants aged 60 and above, recruited through personal networks and through Seniornett Norge, across three testing iterations. The first session used an interactive Figma prototype, the second session a coded prototype, while the final round was conducted on the participants' own devices using the deployed platform, [seniorki.no](https://seniorki.no). A pre/post knowledge test and a post-test System Usability Scale (SUS) were administered in the final iteration to measure changes in understanding and perceived usability.

## 2 LITERATURE REVIEW

This chapter reviews the literature underlying this thesis. Section 2.1 introduces Kaur et al.'s (2025) study, which the platform's curriculum is based on. Section 2.2 covers AI literacy: how it is defined, and the gap in existing resources for older adults. Section 2.3 covers universal design and interface design guidelines for older adults. Section 2.4 covers gamification in learning platforms.

### 2.1 Prior study

This thesis is built upon the previous research of Kaur et al. (2025), who investigated older adults' experiences, motivations, and preferences in learning about AI. The study used a mixed-methods approach: a survey of 182 Norwegian adults aged 60 and above, the same dataset also reported in Kaur and Chen (2023), followed by two rounds of semi-structured interviews with 33 participants in total. The first round focused on learning experiences and motivations, while the second round focused on learning preferences and formats.

Participants were generally motivated to learn about AI, but most had gained what knowledge they had informally, through internet searches, family and friends, and newspapers, rather than any structured instruction (Kaur et al., 2025). Many participants were also concerned about privacy and data sharing (Kaur et al., 2025). Drawing on these findings alongside existing research in AI literacy education, Kaur et al. designed a curriculum intended to give older adults the knowledge and skills needed to understand and effectively use AI-enabled products and services.

This thesis implements that curriculum as an interactive, self-paced online platform, with two notable deviations from the original design: the platform does not include the practical activity specified for each module, a deviation made due to time constraints (see section 4.4), and it is delivered entirely online rather than in the blended format Kaur et al. proposed, which is a deviation driven more by the practical challenges of coordinating in-person sessions as a single researcher than by time itself (see section 5.3.2).



## 2.2 AI literacy

AI literacy is defined as "*a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace*" (Long & Magerko, 2020, p. 2). Ng et al. (2021) organizes this into four components: knowing and understanding AI, using and applying AI, evaluating and creating AI, and understanding the ethical dimensions of AI. These definitions were developed largely with general or younger populations in mind. Kaur and Chen (2023) further extends this concept to older adults, examining what these competencies look like for a demographic that, as established in Chapter 1, often approaches AI with lower trust and higher anxiety than younger users. For the purposes of this thesis, AI literacy is understood in this extended sense: not just the ability to operate specific AI-enabled devices, but the broader knowledge, critical evaluation skills, and confidence needed to engage safely and meaningfully with AI in daily life.

Long and Magerko's definition and Ng et al.'s four-part structure are not competing accounts, but rather different levels of the same idea: Long and Magerko's broader competencies (critically evaluating, communicating and collaborating with, and using AI as a tool) maps loosely onto Ng et al.'s "use and apply" and "evaluate and create" categories, while Ng et al. treats ethics as its own separate category. Long and Magerko also cover ethics, but it's part of their wider list of competencies, not something the core definition itself mentions directly.

Existing AI literacy resources are also not well matched to older adults' needs. Most AI literacy education and research to date have focused on formal settings such as K-12 curriculums, leaving structured AI training in informal, adult learning contexts comparatively limited, and research specifically targeting older adults more limited still (Kaur et al., 2025). This gap remains even in the Norwegian context. A free, Norwegian-language introductory AI course, Elements of AI, is already available (University of Helsinki & MinnaLearn, n.d.), but it targets a broad adult audience rather than older learners, and its content and pace are unlikely to suit a population with lower baseline digital confidence.

Where initiatives do exist for older adults, they tend to focus narrowly on a single AI-enabled product, such as tutorial-based training for voice assistants, which limits their relevance to the broader range of AI technologies older adults are likely to encounter (KangJie et al., 2025). A related gap concerns older adults' awareness of AI-related harms such as misuse, scams, or exploitation, which is rarely addressed directly by existing materials (KangJie et al., 2025). Kaur et al.'s (2025) curriculum, on which this platform is built, addresses this directly through a dedicated module on identifying trustworthy sources, recognizing scams, and understanding AI ethics. This projects platform is designed to address that gap. Rather than teaching a single tool or device, the curriculum on which it is built takes a broader view of AI literacy tailored specifically to the needs, preferences, and challenges identified for older adult learners.

## 2.3 Universal design of ICT and interface design for older adults

This section first outlines universal design as a legal and regulatory requirement in Norway, before turning to empirical findings on interface design specifically for older adults, which together justify the accessibility and interface decisions described in section 4.3.

### 2.3.1 Universal design of ICT

Universal design refers to designing products and environments so that they can be used by everyone to the greatest extent possible, regardless of functional ability (Nordbø, 2017). This is often mistakenly understood as being primarily about blindness, but it also covers a much broader range of conditions, such as color blindness, dyslexia, and situational or permanent physical impairments (Nordbø, 2017). Many of these challenges become more common with age specifically: most people begin needing reading glasses in their forties, and start experiencing difficulty perceiving visual contrast in their sixties (Nordbø, 2017), placing this project's target group within the population universal design is intended to serve. Web accessibility applies these principles specifically to digital content, and the Web Content Accessibility Guidelines (WCAG), developed by the World Wide Web Consortium (W3C), are the primary set of recommendations for how digital content can be made accessible (Nordbø, 2017).

In Norway, digital accessibility is a legal requirement rather than a voluntary best practice. The Forskrift om universell utforming av informasjons- og kommunikasjonsteknologiske (IKT)-løsninger (Regulation on Universal Design of ICT Solutions) has required private sector web solutions to conform to WCAG 2.0 level A and AA since 2013 (Forskrift om universell utforming av IKT-løsninger, 2013). SeniorKI, as a university research project rather than a public or commercial service, is not itself bound by this regulation, but WCAG 2.1 level AA (World Wide Web Consortium [W3C], 2018) was adopted throughout as the platforms accessibility target regardless, both to follow current best practice and because these requirements are directly relevant to a course intended for older adults, a group disproportionately affected by inaccessible design.

### 2.3.2 Interface design for older adults

Research on interface design for older adults has identified recurring usability problems, as well as design choices that help address them. Kaur and Chen (2022) conducted a heuristic evaluation of ten physical and cognitive training apps commonly recommended to older adults by health organizations and ageing-related forums, noting that dedicated apps built specifically for older adults were difficult to find. Using Nielsen's (1994) ten usability heuristics, they found that every app evaluated had usability issues, with the most frequently recurring problems being small font sizes, low contrast, inconsistent interface design, and small touch targets.

Hsieh et al. (2018) approached this from the other direction: in developing and testing a fall-risk mHealth app for older adults, they incorporated large text, high contrast color schemes, multi-modal feedback, and step-by-step instructions from the outset, and found that the resulting app achieved high system usability scale (SUS) scores when tested with older adult participants. Taken together, these two studies support a consistent set of guidelines for this target group:

- Large, high contrast text and UI-elements
- Large touch targets
- Simple and consistent navigation

- Reduced cognitive load through clear, step-by-step instructions

These principles directly informed the accessibility decisions described in section 4.3.1, including the use of relative font sizing (rem) rather than fixed pixels, text labels alongside icon-based controls, large touch targets, and a calm, high contrast visual design.

## 2.4 Gamification in learning platforms

Gamification refers to the use of game design elements in non-game contexts (Deterding et al., 2011). Deterding et al. distinguishes gamification from fully fledged games on the basis that a serious game is designed as a complete game for a non-entertainment purpose, while a gamified application merely borrows specific elements from games, such as challenges, levels, and points, without becoming a game in its own right. This remains the most used definition of gamification in the literature, cited or closely echoed by roughly half of the design frameworks reviewed in a systematic review by Mora et al. (2017).

Research on gamification for older adults specifically remains limited. In a systematic review of twelve peer reviewed studies on gameful interventions for adults aged 55 and older, Koivisto and Malik (2021) found that eleven of the twelve were conducted in the health domain, with adaptive difficulty, social elements, scores and points, clear goals, and progress indicators as the most common gameful elements. Reported effects were mostly positive but weak, and the authors call for further work outside the health domain. This concentration in health applications leaves gamification for older adults in educational contexts relatively unexplored, which mirrors a broader pattern. In their review of gamification design frameworks across domains, Mora et al. (2017) found that frameworks focused specifically on learning made up only 15% of the literature they reviewed, trailing business and generic applications.

One of the few studies to examine gamification for older adults specifically within an e-learning context is Gellner and Buchem (2022), who evaluated a gamification model for senior learners structured around eight motivational “core drivers” adapted from an established gamification design framework. Surveying 41 seniors aged 66 to 93, they found

that intrinsically oriented elements, such as seeing one's own learning progress and receiving instant feedback, were rated the most motivating. Explicitly competitive or extrinsic elements were rated far less favorably. Seniors did not identify with the competition-driven player type, and elements requiring great effort to succeed received negative average ratings. Based on these results, Gellner and Buchem recommended removing quizzes and puzzles from their model, since participants rated them lowest among all techniques tested.

Gellner and Buchem (2022) also found that their sample of seniors only had limited prior experience with digital games and learning applications, despite reporting more familiarity with non-digital games such as cards and puzzles. This distinction between broad enjoyment of games and unfamiliarity with digital gaming conventions specifically supports keeping gamification minimal.

These findings inform the gamification approach taken in this project. Rather than building competitive elements such as leaderboards into SeniorKI, the platform's gamification is limited to three deliberately simple, achievement oriented elements: stars, badges, and a progress overview (see section 4.3.3). This aligns with what Gellner and Buchem's findings suggest resonates with older learners: visible progress and positive feedback, rather than competition or unpredictability.

### 3 METHODOLOGY

This chapter describes the methodology used to develop and evaluate the SeniorKI platform. Section 3.1 outlines the research approach. Section 3.2 describes the development process. Section 3.3 covers the evaluation, including recruitment, and the formative and summative evaluation methods. Section 3.4 describes the qualitative and quantitative data analysis methods used. Section 3.5 addresses ethics and data management.

#### 3.1 Research approach

The project was carried out using a user-centered design (UCD) approach, meaning that members of a target group were involved throughout the design and development process, rather than only being brought in once a finished product was ready for testing. This fit well with a project aimed at older adults, a group whose digital literacy and needs vary a lot (see Chapter 1) and aren't always well represented in typical software development (see section 2.3.2). According to Preece et al. (2019), Gould and Lewis (1985) laid down three principles that underpin a user-centered design approach: an early focus on users and their tasks, empirical measurement of how users respond to designs and prototypes, and iterative design, where problems found during testing are fixed and then tested again. All three were reflected in this project, which involved participants across two iterations of the platform before a more formal summative evaluation of the third iteration.

The project combined qualitative and quantitative methods. During the two formative iterations, data collection was mainly qualitative, with think aloud observations and open coding of the issues participants ran into while using the prototype. For the final summative evaluation, a quantitative element was added, in the form of a pre/post knowledge test and a System Usability Scale (SUS) questionnaire, alongside open-ended questions that were analyzed qualitatively. Combining these approaches made it possible to refine the platform's design early on, and later measure whether it actually had an effect on participants' AI understanding, alongside how usable it was perceived to be.

## 3.2 Development process

The prototype was developed across three main stages: paper prototyping, Figma-based digital prototype, and a coded React implementation deployed as a functioning website. Each stage built upon the previous stage.

Although research shows that most older adults in Norway now access the internet via a smartphone, 95% of those aged 65-74 had done so within the past three months as of 2025 (Statistics Norway [SSB], 2025), a mobile-first design approach was not adopted for this platform. Because the platform's content is primarily text-based, consisting of reading material rather than short interactions, I assumed that a larger screen in landscape orientation would offer a more comfortable reading experience than a mobile phone, and designed primarily for that context. Mobile phone use was treated as a secondary consideration rather than the primary design target. To still accommodate participants without access to a larger device, the platform was designed to be responsive, so that it remained usable on a mobile phone as well in its final iteration.

### Paper prototyping

The first stage consisted of drawing paper sketches exploring several layout options for the platform's key screens. Low fidelity prototypes such as these are quick and cheap to make, which makes them well suited to exploring and discarding ideas early in the design process, before committing to an idea (Nordbø, 2017). For this project, several sketch variations of the main course screens were drawn and compared, and elements from the preferred sketches were carried forward into the next stage.

The initial project plan, suggested by the supervisor, included a usability test with paper prototypes involving two participants, but this was not carried out. Nordbø (2017) points out that low-fidelity prototypes can be difficult for people unfamiliar with prototyping to interpret, since they may assume the prototype reflects what the finished product will actually look like, and be disappointed when it falls short of that expectation. Given that the target group for this project can be assumed to have limited prior experience with user interface prototypes, testing directly on paper sketches risked producing exactly this kind of

confusion, rather than useful feedback on the specific layout ideas. The first usability test was therefore conducted on the Figma prototype instead, described below.

## Figma design

Elements from the preferred paper sketches were used to create the mid-fidelity, clickable digital prototype using Figma. This stage established the visual language of the platform, including its color palette, typography, spacing, and the design of reusable components like buttons, cards, and progress indicators. Because this prototype was interactive and visually closer to a finished product, it was judged to be better suited than the paper sketches to usability testing with the target group. This prototype was used in the first round of user testing, and it was refined after each of the two formative usability tests.

## React implementation

The coded prototype for the second iteration was implemented as a React web application, using TypeScript and Vite for the build tooling. React was chosen for its component based architecture, which supported the platform's iterative design process, and its wide adoption and documentation, which made it a practical choice for AI-assisted development. In the third iteration, this code (see Appendix B) was deployed via GitHub Pages under a custom domain (seniorki.no). This made it possible to test real interactions rather than a partly clickable, but ultimately limited design.

Claude Code, an AI coding agent, was used to help implement this stage of the platform. Although React was the technology I was most experienced with, coding speed remained a limiting factor for how much I could realistically implement within the project's timeframe. Using Claude Code was expected to increase implementation speed and code quality beyond what would have been achievable through manual coding alone, making it possible to implement larger design changes without a proportional increase in development time. My existing familiarity with React and TypeScript meant I could still make manual contributions or corrections when needed, rather than being fully dependent on AI-generated code. Claude Code was also used to implement functionality that would otherwise have been difficult to complete within the available time, most notably responsive design across a



range of screen sizes, which is necessary to accommodate the variety of personal devices participants might use during the final summative evaluation, as opposed to the single tablet I provided in the earlier formative usability tests.

### Curriculum content development

I wrote the initial content for each module based primarily on articles from Store Norske Leksikon (snl.no) and popular sources where specific examples were needed, with the target group in mind regarding language, tone, and technical level (see Appendix G). The original plan was to have this content reviewed by Seniornett course leaders to assess whether its complexity and tone were appropriate for the target group, drawing on their direct experience with its population. Due to time constraints, this review did not take place as planned. As a fallback, an AI large-language model (Claude) was used instead, and given a detailed prompt (see Appendix J) describing the target group and the intended tone. This final version (see Appendix H) was proofread and approved by me and added as the course's content in the code.

## 3.3 Evaluation

The platform was evaluated in two stages: two formative usability testing iterations conducted to guide design decisions, and a final summative evaluation of the deployed platform.

### 3.3.1 Recruitment

Participants were recruited through two channels: Seniornett Norge (specifically its Sarpsborg Øst chapter) and my personal and family network. Recruitment from Seniornett was preceded by, and alongside, approximately ten weeks of volunteer work at the chapter's weekly datacafé sessions, where I assisted members with general digital device use. This volunteering served two purposes: it built familiarity and trust with members of the target group before recruiting, and it gave me informal, firsthand exposure to the kinds of difficulties older adults commonly encounter with digital devices, which was informative when designing the early iterations of the platform.

This approach differs from a single contact convenience sample in that I establish an informal relationship with possible participants before recruitment begins. The initial recruitment pool, from both Seniornett and family, was broader and included a number of Seniornett members with little or no prior personal contact with me.

Across the two formative iterations and the final summative evaluation, a total of 14 participants took part in testing the platform, and a total of 17 tests were performed, since three of the participants took part in both a formative and the summative evaluation. Two of these 14, both non-completers in the summative evaluation, did not provide data used in this thesis and are discussed separately as attrition in sections 3.3.3 and 5.2.4. The demographic information of the remaining 12 are given below. The recruitment pool for the summative evaluation in particular, was broader than the group that went on to complete it. This is discussed further in sections 3.3.3 and 5.2.4.

For all 12 unique participants who took part in the formative and summative evaluations, the sample was balanced by gender (6 female, 6 male) and ranged in age from 60 to 83. Education levels varied: 3 participants had completed high school (videregående skole), 4 held vocational qualification (fagbrev or yrkesutdanning), and 5 held a college or university degree. This composition is discussed further in section 5.1.1.

### 3.3.2 Formative evaluation

Two rounds of formative usability testing were conducted before the final summative evaluation. The first round testing the Figma prototype, and the second round testing the coded React prototype. Both rounds used a Samsung Galaxy Tab A9 tablet that I provided, held in landscape orientation, to keep the testing environment consistent across iterations. Sessions lasted approximately 10 minutes each and took place either at Seniornett's premises or in family members' homes, depending on the participant.

According to Nielsen (2000), testing with five participants is sufficient to identify around 85% of usability problems in an interface, and running several smaller tests across iterations

tends to be more effective than one large test. Five participants was therefore set as the target for each formative round. This target was met in the first iteration, which had five participants, but not in the second, which had four planned, but due to a last minute cancellation, ended up with only three participants. This shortfall is acknowledged as a limitation in section 5.2.2, since testing with fewer than the recommended number increases the risk that some usability problems in the coded prototype went undetected before the platform was tested more thoroughly in the final evaluation.

In the first iteration, participants completed 18 short tasks covering seven areas of the platform: the welcome screen, the course overview, the progress overview, the chapter overview, individual topic pages, the quiz, and the badge earned screen. Each task asked the participant to carry out a specific action, such as returning to the course overview or starting the next chapter, while thinking aloud about what they saw and expected to happen. The session closed with three brief reflection questions about the experience as a whole. The evaluation protocol for Iteration 1 can be seen in Appendix C.

The second iteration used a similar structure of 18 tasks, but was not entirely identical to the first. Two tasks introducing a new onboarding flow were added at the start of the session, directly addressing confusion about the platform's underlying concept observed in the first iteration, and a task testing a new text-size control was added, following a request from a participant in the first iteration for larger text. Most other tasks were carried over unchanged from the first iteration, since repeating the same task allowed it to test whether a specific design change had resolved the issue it was meant to address, for example whether the new floating buttons resolved the earlier problem of users forgetting what was available at the top of the page after scrolling down. Task outcomes were not coded against a success or failure criteria: see section 3.4.1 for how observations were analyzed, and Appendix D for the evaluation protocol for Iteration 2.

I ran all the testing sessions alone, without audio or video recording. This decision was informed by earlier project experience with older users suggesting that visual observation of how participants interacted with the interface, like hesitation, incorrect taps, backtracking, confusion about where to find something, was the most valuable source of usability

information, more so than the content of verbal think-aloud commentary itself. Recording equipment was also avoided out of concern that it would make an already unfamiliar testing situation feel more intrusive for a target group with limited prior exposure to this kind of evaluation. This concern proved reasonable, as even without recording, participants showed visible signs of discomfort to varying degrees during testing, an observation discussed further in section 5.2.6.

During each session I took minimal notes, in order to keep the interaction as relaxed as possible and avoid making participants feel closely observed or scrutinized while completing the tasks. Fuller notes were written up immediately after each session ended, while details were still fresh in mind. This approach traded some completeness and verbatim accuracy, particularly of the participants' spoken commentary, for a more comfortable testing experience, and it means the qualitative analysis in this thesis draws primarily on reconstructed behavioral observations rather than transcribed dialogue.

Issues identified during each round were noted and used to inform design changes before the next iteration, described further in section 4.2.

### 3.3.3 Summative evaluation

The summative evaluation measured the effect of the finished, deployed platform on participants' AI understanding and its perceived usability. Unlike the formative rounds, which relied on observation during short, moderated sessions, this evaluation used a pre/post knowledge questionnaire, a post-use usability questionnaire, and open-ended questions over an extended period of independent use, allowing quantitative comparison alongside qualitative feedback from a few open ended questions.

Participants were recruited through the same channels described in section 3.3.1. 15 people signed up to participate. 9 began the study by completing the pre-test, and 7 completed both the pre- and post-test, forming the matched pair sample used in the knowledge test analysis (see section 3.4). Of these 7, 3 had also participated in one of the earlier formative iterations (1 from iteration 1, 2 from iteration 2). The gap between the initial recruitment

pool and the number who completed the study is discussed further as a limitation in section 5.2.4.

Participants used the platform independently over a period of up to two weeks between the pre- and post-test, on their own personal devices rather than the single tablet used in the formative rounds.

Data collection used three instruments, distributed through Nettskjema (see section 1.3 and Appendix E and F):

- A knowledge test, administered both before and after the two-week usage period, consisting of 10 multiple choice questions covering concepts taught across the course. Questions were deliberately worded differently from the in-course quiz questions, so that the test measured retained and transferable understanding of the material rather than recall of quiz phrasing.
- A System Usability Scale (SUS) questionnaire, used to measure participants' perceived usability of the platform.
- A set of open-ended questions following the SUS questionnaire, asking participants to describe their experience with the platform in their own words, analyzed qualitatively (see section 3.4). These questions were the only ones that did not require obligatory answers from the participants to complete the form.

An alternative, more recent usability instrument, developed by Zaharias and Poylymenakou (2009) specifically for e-learning applications, was considered on my supervisor's recommendation before SUS was chosen. Their questionnaire extends usability measurement beyond functional usability, to also include a measure of learner motivation and engagement, arguing that traditional usability instruments miss this aspect in e-learning contexts. However, its final version consists of 49 items across eight usability and motivation parameters, which is considerably longer than SUS's 10 items, and it was validated on corporate employees that were largely IT-professionals, rather than on older adults with limited digital competence. Given the length of the instrument and the different population it was developed and validated for, I judged it unsuitable for this project's target group. Age-related decline in cognitive abilities is well documented (Kaur & Chen, 2022), and a

substantially longer questionnaire risked increasing cognitive load and fatigue for participants beyond what the study's post-test already asked of them, alongside the 10-question knowledge test and open-ended questions.

## 3.4 Data analysis

### 3.4.1 Qualitative data analysis

For all three iterations, the observation notes and open-text feedback were grouped into informal categories such as gamification issues, navigation issues, and layout issues, based on a few read-throughs of the data. Task-level difficulties, such as hesitation, backtracking, or needing a hint, were noted together with the same categories rather than coded against a predefined success or failure criteria for each task. This process drew on the general logic of thematic analysis, in which patterns are identified and grouped into categories to help make sense of qualitative data (Braun & Clarke, 2006, as cited in Preece et al., 2019), but did not follow its full procedure: categories were assigned directly during the review by a single researcher, rather than through a separate, more systematic coding pass with a second coder to check agreement.

This approach is therefore better described as informal, category-based grouping inspired by thematic analysis, rather than a full application of it. This was a practical choice given the scope of a single researcher master thesis: with no second coder available, a full multi-phase, independently verified coding process was not feasible within the project's timeframe. This is acknowledged as a limitation in section 5.2.5, since interpretations of qualitative data reflect a single researcher's reading rather than one checked for agreement against an independent perspective.

Once grouped, categories from the formative rounds were reviewed to separate the issues pointing towards concrete design changes, from lower priority issues that was either not tested thoroughly enough to act on with confidence, or issues not judged to meaningfully affect usability.

### 3.4.2 Quantitative data analysis

Quantitative data from the knowledge test were analyzed using a Wilcoxon signed-rank test, comparing pre- and post-test scores for the 7 participants who completed both tests. This non-parametric test ranks the size of each pair's difference rather than assuming those differences are normally distributed, and was chosen over a paired t-test because the pre/post differences were small in number and unevenly distributed, clustering toward a ceiling at the maximum possible score. This pattern made the normality assumption underlying a paired t-test difficult to justify. A sign test was also considered, since it requires fewer assumptions than the Wilcoxon test, but it discards the magnitude of each participant's change and only considers its direction. Given that the size of individual improvements, ranging from 1 to 5 points, is informative in itself, the sign test was judged to discard information the Wilcoxon test could retain, and was therefore not used as the primary test.

SUS responses were scored following the standard procedure, converting the 10-item questionnaire into a single score from 0 to 100 per participant. Given the small sample size, SUS scores were only reported as individual and average scores, rather than testing for statistical significance. This mirrors the approach taken with the knowledge test, where the small effective sample also limited the scope of quantitative claims that could be supported. Scores were compared against benchmarks from Bangor, Kortum, and Miller (2008), and a sub-scale breakdown, following Lewis and Sauro (2009), was used to examine usability and learnability separately, alongside a descriptive review of individual item scores.

## 3.5 Ethics and data management

Ethical considerations relevant to this project, including SIKT approval, informed consent procedures, data storage and retention, and pseudonymization of participants, are addressed in section 1.3. This includes the handling of personal data collected during the summative evaluation, where participants' names and email addresses were temporarily stored to link pre- and post-test responses, and the decision not to record formative testing sessions in order to reduce discomfort for participants unfamiliar with this kind of evaluation.

## 4 RESULTS

This chapter presents the SeniorKI platform and the results of its development and evaluation. Section 4.1 gives an overview of the platform. Section 4.2 covers the two formative usability testing iterations and the design changes that followed from each, summarized in a table in section 4.2.3. Section 4.3 describes the final platform design, covering accessibility, visual design, and gamification elements. Section 4.4 describes the curriculum structure implemented in the platform. Section 4.5 presents the results of the summative evaluation, covering the quantitative knowledge test and SUS results, and the qualitative open-ended feedback.

### 4.1 Platform overview

The platform is a self-paced, web-based AI literacy course designed for Norwegian adults aged 60 and above. The platform is accessible via web browser and does not require any installation. It is also responsive, to be usable on a number of different devices and screen sizes.

The main goal of the platform is to give users a practical, non-intimidating introduction to AI. It will not make them technical experts, but they will gain enough knowledge and skills to know the basics of what AI is, and the ability to navigate AI-enabled products and services in their daily lives safely and with more confidence.

The platform is structured around five learning modules, each containing a few short lessons. Each lesson is designed to be completed in around 1-5 minutes, with further reading about a topic provided, but optional. Each module includes theoretical explanations and a short quiz covering the modules topics. No additional interactive activities beyond the quizzes were implemented, due to time constraints (discussed further as future work in section 5.3.1). Progress through the platform is visualized via a map-style overview screen showing completed and upcoming lessons, and users can earn badges by completing each module, and stars by completing quizzes.



### 4.1.1 Platform presentation

#### Welcome

This is the first page a visitor sees at [seniorki.no](https://seniorki.no). It contains a friendly illustration showing an older man speaking with a robot, with a short value proposition and a call to action.

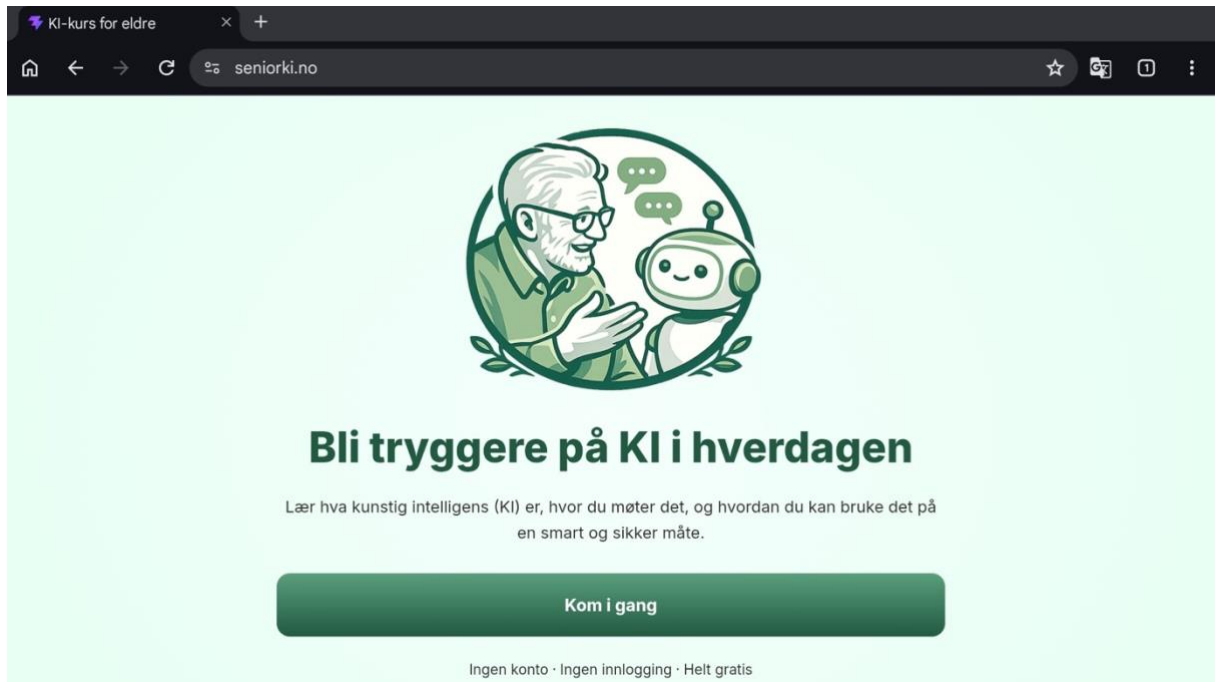


Figure 4.1: The [seniorki.no](https://seniorki.no) welcome screen, shown on an Android tablet in landscape orientation.

The headline “Bli tryggere på KI i hverdagen” (Become more confident with AI in everyday life), and the call to action beneath it states the course’s purpose directly: what AI is, where it shows up, and how to use it safely. Below that, a single large button, “Kom i gang” (Get started), is the only interactive element on the page, and the reassurance text beneath it, “Ingen konto, ingen innlogging, helt gratis” (No account, no login, completely free), removes friction that could otherwise make hesitant users avoid continuing.

The single-button layout is a direct outcome of Iteration 1 testing (see section 4.2.1), where the original two-button welcome screen (“Get started” / “Continue where I left off”) caused three out of five participants to tap through without noticing there were two options at all.

Since the second button was only relevant on a return visit, it was removed from the screen entirely.

## Onboarding / Help

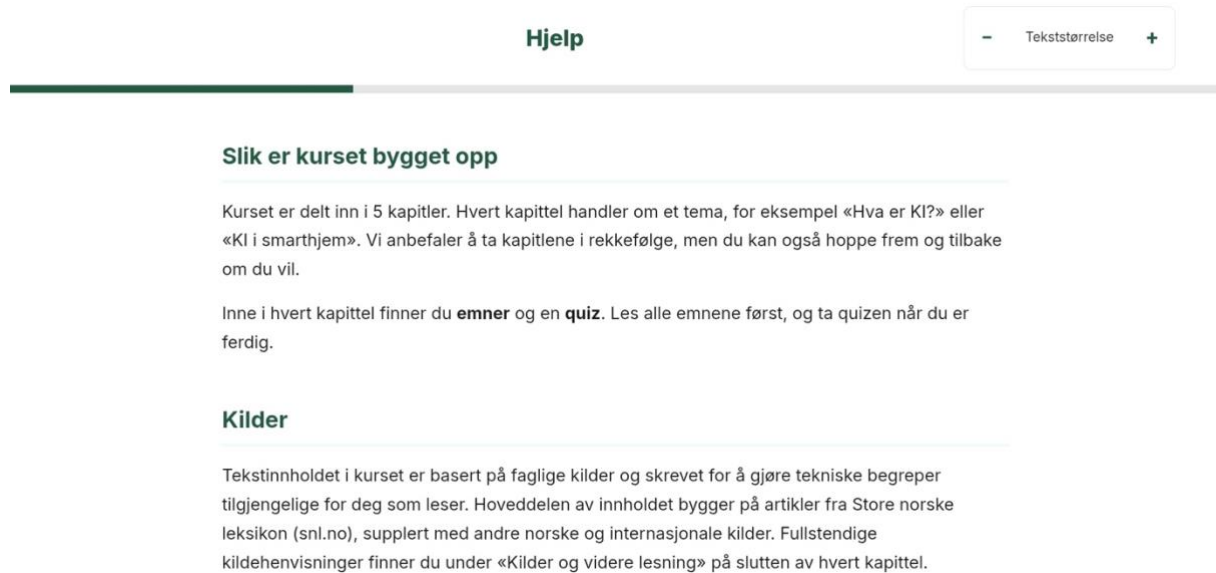


Figure 4.2: The onboarding/help page, upper portion, showing the course structure explanation and content sourcing.

Following the welcome page, users are shown an onboarding page explaining how the course works before starting. It describes the five-module structure, how to answer quizzes, the star and badge system, how to navigate around the platform, and how to use the text size button. Finally, the page mentions that it remains accessible at any later point by pressing the "Help"-button, which is available on most pages. At the bottom of the screen is a button that either starts the course or navigates back to the course, depending on whether the users came from the welcome page, in which case the button text says "Trykk her for å starte kurset" (Press here to start the course), or from somewhere in the course via the info-button, where the button will say "Fortsett kurset" (Continue the course).

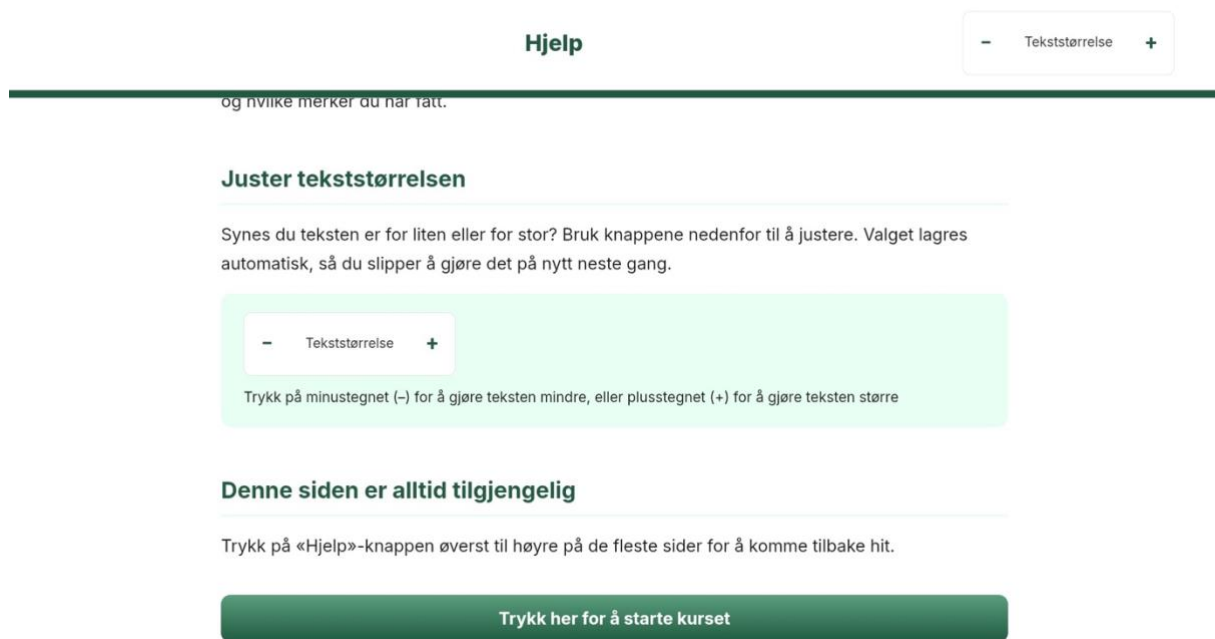


Figure 4.3: The onboarding/help page, lower portion, showing the text-size control and the button to begin or continue the course.

The information page was added in Iteration 2, in response to Iteration 1 findings. Two participants reported being unsure about the platform's underlying concept, and this onboarding step was introduced to address that confusion before a user begins the course itself.

### Module overview

This page shows the six chapter cards in a two-column grid, alongside a header summarizing overall progress and three buttons at the top: the text size adjustment, a "Se fremgang (See progress) button, and the "Hjelp" (Help) button linking back to the onboarding page.

In-progress and not yet started modules use a card style with a white background and an outline to distinguish them from the completed modules (1 and 2 on Figure 4.4), which are shown with a filled dark green background and display the stars earned from that chapter's quiz. One can also see that "stars earned" and "parts completed" are tracked separately, as Module 1 shows all three parts read and that two stars are earned, but Module 3 shows that two out of three parts are read and no stars are earned. For a module to be marked as

finished, all topics must be read and the quiz must be completed with at least one correct answer. Progress on in-progress modules are shown on the cards with gray text.

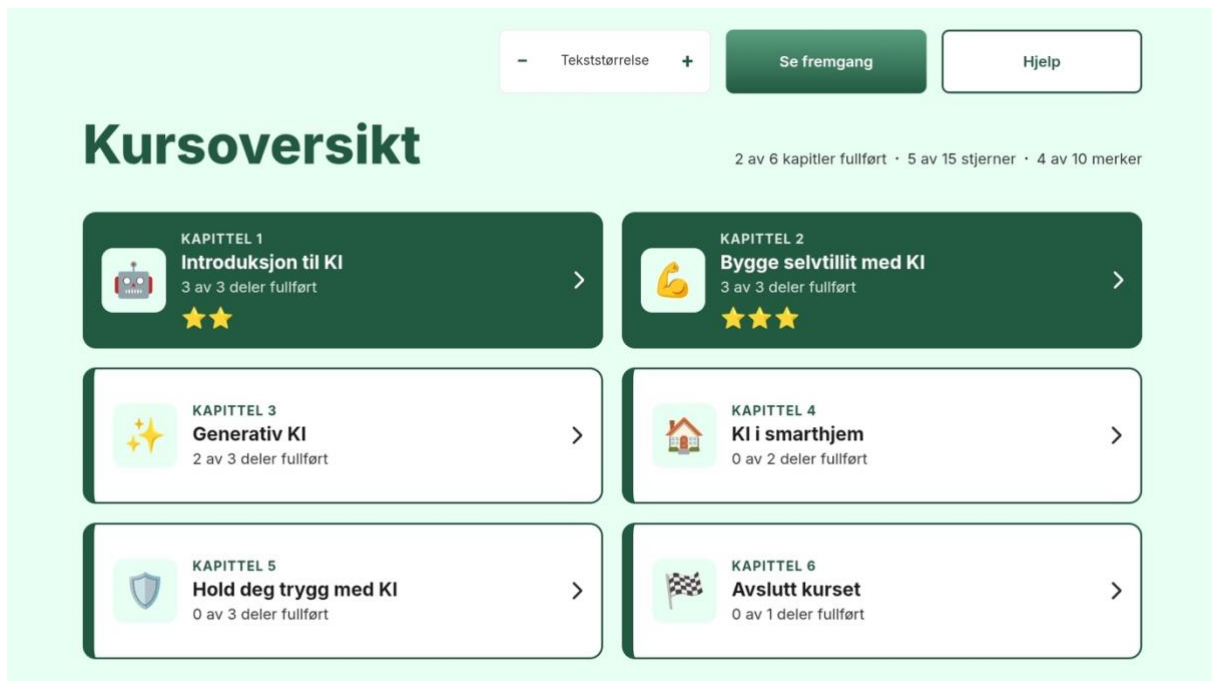


Figure 4.4: The course overview page, showing six modules, with two of them completed, and a different number of stars earned.

The header text “2 av 6 kapitler fullført · 5 av 15 stjerner · 4 av 10 merker” (2 of 6 chapters completed · 5 of 15 stars · 4 of 10 badges) is the brief text summary that replaced the earlier bullet point topic overview on this page, following Iteration 1 testing (see section 4.2.1). The per module progress bar from Iteration 1, which testing found was not noticed by any participant and functioned as visual clutter was also removed from this page. The visual progress information was instead moved to the “Se fremgang” (See progress) page, and presented there as a list.

Removing the progress bar freed up space within each card, which made it possible to arrange the modules in a two-column grid rather than a single full-width column. A sixth chapter, “Avslutt kurset” (Finish the course), was added at this point as well. This was both because six modules divide evenly into a two-column layout, unlike the original five, and because it gives participants a clear final step for navigating to the post-test Nettskjema. This chapter is temporary, added specifically for the study period rather than as a permanent

part of the platform: it consists of a thank-you message for participating, a direct link to the post-test Nettskjema, and a book recommendation: *Maskiner som tenker* by Inga Strømke, for users who want a more advanced explanation of AI than the course itself provides.

The new grid layout meant that all six modules could fit on the screen without scrolling, which was a deliberate response to Iteration 1 testing, where the issue was not scrolling itself, but that all five participants who scrolled down forgot what was no longer visible at the top of the screen, such as the “See progress” button. Removing the need to scroll on this page, together with the text size, progress, and help buttons remaining on the top on all pages, regardless of scroll position, directly addresses this. The same two-column, non-scrolling approach was also applied to the topic page, covered next in this walkthrough.

### Topic overview

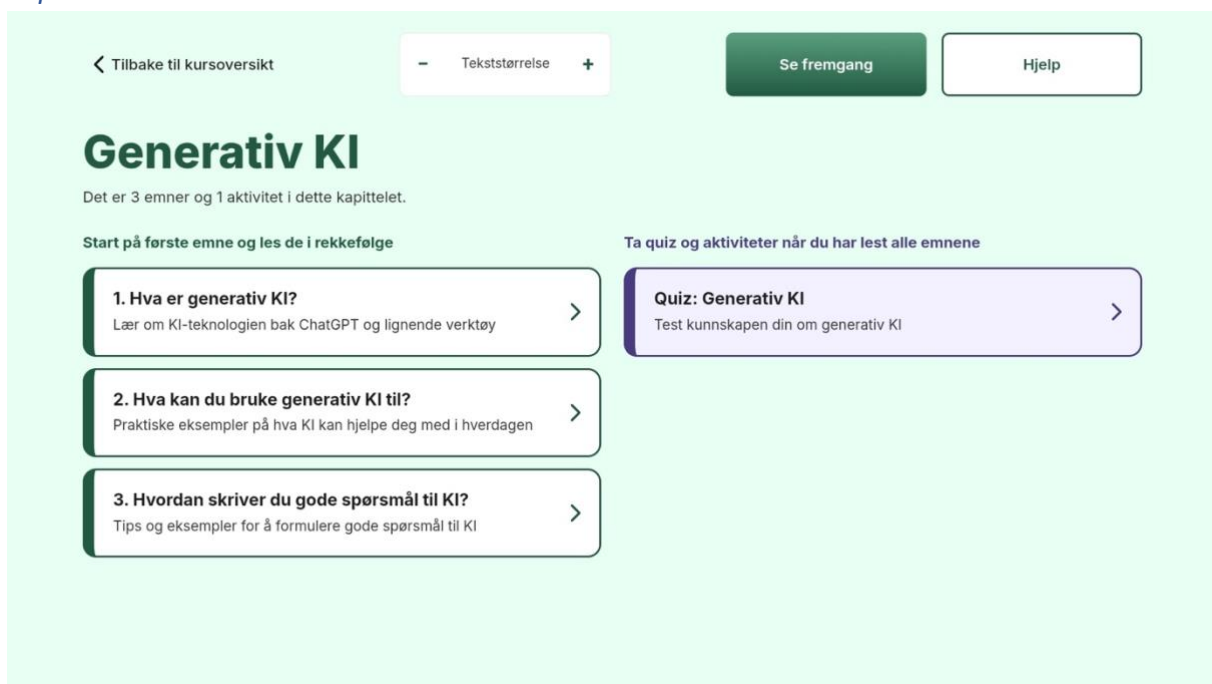


Figure 4.5: The topic overview page for the "Generativ KI" chapter, showing three numbered topic cards and a separated quiz card.

This page splits theory topics and the quiz into two visually and spatially distinct columns. The left column has the heading “Start på første emne og les de i rekkefølge” (Start with the first topic and read them in order), and lists the module’s three topics as numbered cards with green outline and accent stripe. The right column, with the heading "Ta quiz og

aktiviteter når du har lest alle emnene" (Take the quiz and activities once you've read all the topics), contains a single, purple colored card for the chapter's quiz. This gives the topic/quiz distinction three different cues at once: position (left vs right), color (green vs purple), and text explaining each element, rather than relying on any single one.

The right-hand column's heading reflects that the quiz is treated as one of potentially several different activities. Currently, the quiz is the only implemented activity in all of the five modules, but this wasn't always the plan, which is discussed below, and it opens possibilities for additions in the future.

This page's layout is the result of two rounds of change following usability testing, covered in full in section 4.2. In the first iteration, topics and activities were listed in a single, full-width column, with activities placed below the topics and requiring the user to scroll down to reach them. This iteration, and the next, also included a second, not implemented placeholder activity on writing effective AI prompts, shown to participants to demonstrate the intended layout before the course content itself existed. In the second iteration, the layout was changed so that the full topic and activity list fit on screen without scrolling. In the third and final iteration, topic numbering and the explicit "read topics in order, then take the quiz" heading text was added, in response to participants appearing hesitant about task order and how to proceed after entering the topic overview during the second round of testing. The placeholder prompt writing activity was not developed further as a separate activity, but the underlying idea was instead incorporated as reading content, becoming Topic 3 of this module: "Ta quiz og aktiviteter når du har lest alle emnene" (Take the quiz and activities once you've read all the topics).

### *Topic content page*

This page displays the reading content for an individual topic within a chapter, laid out as body text below the persistent header buttons (text size, progress, help) already introduced earlier in this walkthrough. The topic's title is shown alongside its position within the chapter ("Emne 2 av 3") and an estimated reading time, calculated from word count. Navigation at the bottom of the page ("Forrige emne" / "Neste emne") allows the user to navigate

backwards and forwards between topics within the module. The content itself is discussed in section 4.4.

The screenshot shows a user interface for a topic page. At the top, there is a navigation bar with a back button labeled 'Tilbake til emneoversikt', a text size control with minus and plus icons and the label 'Tekststørrelse', a green button labeled 'Se fremgang', and a button labeled 'Hjelp'. Below the navigation bar, the main content area has a title 'Hva kan du bruke generativ KI til?' in a large, bold, dark green font. Under the title, it says 'Emne 2 av 3' on the left and 'Ca. 4 min' on the right. The body text starts with a paragraph: 'Generativ KI er lett å bruke, og mulighetene er mange. Her er noen konkrete eksempler på hva som kan være mulig.' This is followed by a section header 'Få svar på tekniske spørsmål' and two paragraphs of text. At the bottom of the page, there are two buttons: 'Forrige emne' on the left and 'Neste emne >' on the right.

Figure 4.6: A topic content page, showing the topic title, position within the chapter, estimated reading time, and body text.

This screenshot shows the same topic page as Figure 4.6, but scrolled down to the bottom. The navigation bar at the top is identical. The main content area now shows a list of input fields: '[Navn]', '[Adresse]', '[Telefonnummer]', and '[E-post]'. Below these is a paragraph of text: 'Hvis svaret ikke er godt nok etter første forsøk, kan du gi flere detaljer og be om spesifikk endringer i stedet for å starte helt på nytt.' This is followed by another paragraph: 'Unngå å skrive inn sensitive opplysninger som personnummer, passord eller bankopplysninger. Slik informasjon har KI-verktøyet ikke bruk for, og du bør aldri dele det digitalt på denne måten.' Then, there is a section header 'Hobbybruk' and a paragraph: 'Generativ KI kan gi råd og hjelp i hobbysammenheng – for eksempel i hagen. Hvis du har et blomsterbed der planter ikke ser ut til å trives, kan du skrive inn hvor i landet du bor, hva slags solforhold det er i bedet, hva slags jord, og annen relevant informasjon, og spørre om hva som kan passe. Eksempler på gode og dårlige spørsmål om dette finner du i neste emne.' At the bottom, the 'Forrige emne' button is still on the left, but the 'Neste emne >' button on the right is now a solid green button.

Figure 4.7: The same topic page, scrolled to the bottom, showing the "next topic" button has become available and the scrollbar has been filled.

Two elements on this page were added after iteration 2, addressing a concern I noted at the time about making sure users were actually aware that there may be content further down the page, so they read a topic's full content before pressing the next-button, which was always active and available. To make this clearer, I added a thin progress bar beneath the header to indicate how much of the current page remains to be scrolled, which fills up further as less scrolling remains on the current page. I tried both vertical and a horizontal version of this indicator during redesign and chose the horizontal version, which I found more visible. The "Neste emne" (Next topic) button is also made inactive until the user has scrolled to the bottom of the page, shown in Figure 4.6 as inactive since the page has not been fully scrolled, and in Figure 4.7 as active because it is fully scrolled. I had originally also considered changing the button text while inactive to "Read the text before continuing", but once the button was made inactive until the page was fully scrolled, I saw this as no longer necessary, as the scrolling requirement itself ensured that the text had to be read to proceed.

### Quiz page

[← Tilbake til kapitteloversikt](#)

## Quiz - Kapittel 1: Introduksjon til KI

2/3

**Hva er nevrale nettverk?**  
Velg ett riktig svar

|                                                         |                                                                             |
|---------------------------------------------------------|-----------------------------------------------------------------------------|
| Et nettverk av datamaskiner som er koblet sammen        | Programvare som simulerer de elektriske signalene i hjernen nøyaktig        |
| Et system som bruker faste regler for å ta beslutninger | En avansert form for maskinlæring, løst inspirert av nervecellene i hjernen |

Sjekk svar

Figure 4.8: Quiz page with an answer option selected, but not confirmed.



Each module ends with a short multiple choice quiz, shown as a 2x2 grid of answer options beneath the question text, with a stepper at the top indicating the current question number out of the total for the quiz. Selecting an option highlights it, changing the card background from light purple to dark purple, without confirming the answer. Pressing “Sjekk svar” (Check answer) then locks an answer and reveals whether the selection was correct.

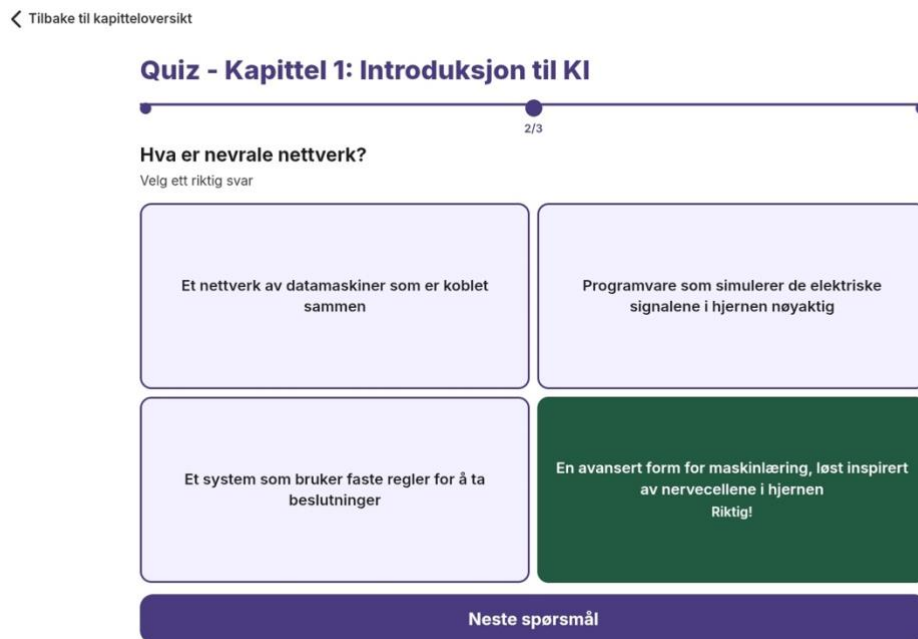


Figure 4.9: Quiz feedback showing a correct answer, with the selected card turned green and labeled "Riktig!"

Feedback is shown directly on the selected card. A correct answer turns the card green with the word “Riktig!” (Correct!) added beneath the answer text, while an incorrect answer turns the selected card red with “Feil” (Wrong), and also highlighting the correct option in green with the label “Dette er det riktige svaret” (This is the correct answer). This shows the user their mistake and the right answer at the same time without requiring a second interaction. Pairing each color with a text label, rather than relying on red/green colors alone, was a deliberate accessibility choice to remain legible for colorblind users. Depending on whether more questions remain, the button below then reads either “Neste spørsmål” (Next question) or “Se resultat” (See result).

The feedback design changed after Iteration 1. The original version introduced a separate feedback panel next to the answer grid after confirming an answer, causing the page’s layout

to shift and elements to move position. This was found to be confusing and was replaced with the current design, where feedback appears on the existing cards without changing the page layout.

[← Tilbake til kapitteloversikt](#)

### Quiz - Kapittel 1: Introduksjon til KI

3/3

Hva menes med at et KI-verktøy som ChatGPT «hallusinerer»?

Velg ett riktig svar

At KI-verktøyet tror det er bevisst og har egne meninger  
Feil

At KI-verktøyet nekter å svare på vanskelige spørsmål

At KI-verktøyet produserer svar som høres troverdige ut, men inneholder feil  
Dette er det riktige svaret

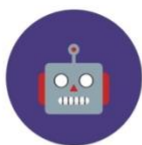
At KI-verktøyet svarer raskt uten å tenke seg om

Se resultat

Figure 4.10: Quiz feedback showing an incorrect answer, with both the user's selection and the correct answer highlighted.

### Post quiz feedback page

[← Tilbake til kapitteloversikt](#)



## Godt jobbet!

Quiz for kapittel 1: Introduksjon til KI

★★☆ 2 av 3 riktige



2 NYE MERKER OPPTJENT!  
Grundig!



Gå gjennom svarene dine



SPØRSMÅL 1: RIKTIG

Hva kjennetegner maskinlæring?

Ditt svar: Det lærer av eksempler og forbedrer seg over tid



SPØRSMÅL 2: RIKTIG

Hva er nevrale nettverk?

Figure 4.11: The post quiz results page, upper portion, showing the star rating and newly earned badges.

After submitting all answers in a quiz, users see a results page summarizing their performance: a star rating (one star per correct answer), followed by a per question breakdown. Each question is shown with a “Riktig”/”Feil” (correct/incorrect) status, the user’s own answer, and for incorrect answers, the correct answer is shown directly beneath it. Incorrect answers also include a “Les emnet på nytt” (Re-read the topic) button, which should link back to the specific topic page, but at the time of delivery simply links back to the topic overview page. Two options are given at the bottom of the page: retry the quiz or continue to the rest of the course.



Figure 4.12: The post quiz results page, lower portion, showing the per question answer review.

This detailed review was not present in the first iteration, which only showed the overall star rating with no per-question feedback. It was added in Iteration 2, allowing users to see specifically which answers they got wrong and to revisit the relevant topic content directly, rather than only knowing their overall score.

### Badge earned page



Figure 4.13: The badge earned page, showing a newly awarded badge.

When a user earns one or more badges, for example by improving a quiz result, which awards the “Enda bedre! (Even better!)” badge, a dedicated page presents the badge before returning the user to the course. The page shows a “Nytt merke” (New badge) label, the badge’s icon and name, a short description of how it was earned, and the total number of badges earned. If more than one badge was earned at the same time, the page steps through them one at a time via a “Neste merke” (Next badge) button. Once all newly earned badges have been shown, the user continues by the same button, which now is labeled “Fortsett kurset” (Continue the course). The user can also choose to navigate to the progress overview, by pressing the “Se alle merkene mine” (See all my badges) button.

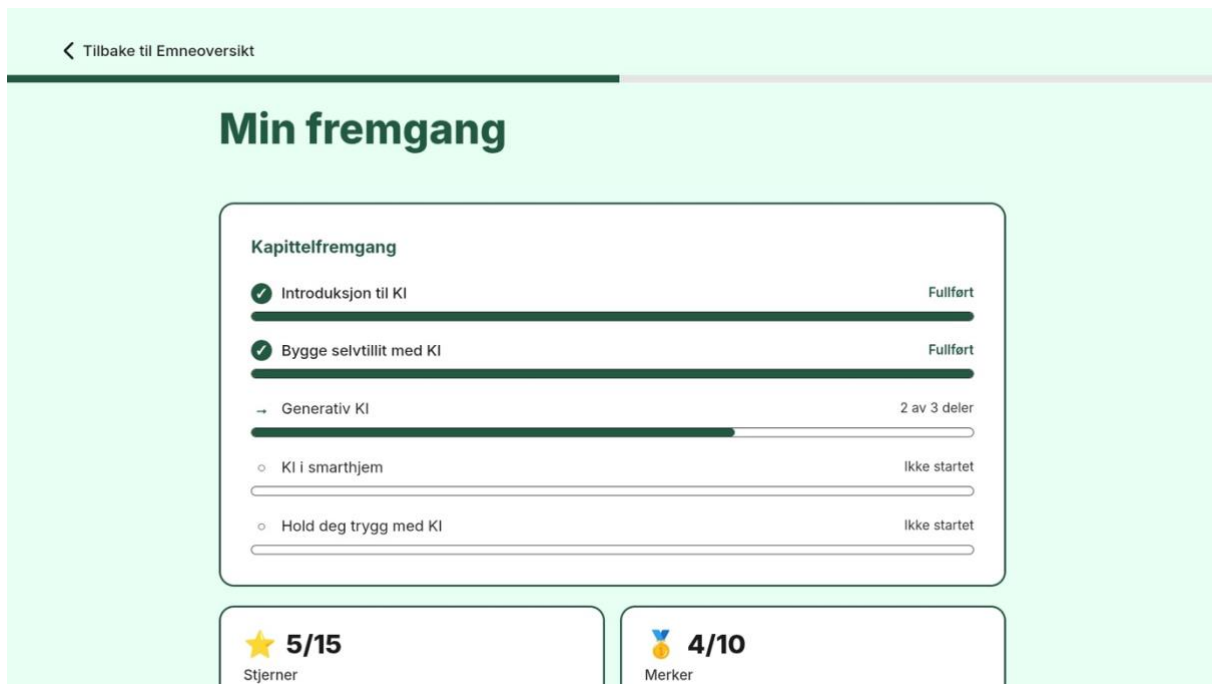


Figure 4.14: The progress page, upper portion, showing per-chapter progress bars.

This page, reached via the persistent “Se fremgang” button available across the platform, gives a fuller view of the user’s progress than the brief text summary on the module overview page. The back-navigation button on the top left of the page is context dependent, returning the user to whichever page they navigated from, rather than a fixed page. A module progress section lists each module individually with its own progress bar and status (completed, in-progress, and not yet started). Below this, total star and badge counts are shown, followed by the full set of available badges, split into “Kapittelmerker” (Module badges, one per module, earned by completing it) and “Spesialmerker” (Special badges, earned through repetition or improvement). Earned badges are shown in full color with an “Oppnådd” (Achieved) label, and unearned badges are shown in a muted style with the requirement needed to unlock them.

This per-chapter progress bar list was added to this page after Iteration 1, when the horizontal stepper previously shown at the top of the module overview and topic overview pages was removed for visual consistency (see section 4.2.1). Rather than dropping progress visualization entirely, this functionality was moved here, keeping the module overview and topic overview pages visually simpler while still giving users who are interested in tracking

their progress a dedicated place to see it. The badge icons were also redesigned at this point, replacing the earlier illustrated robot mascots with simpler icons. This addressed a legibility concern with the responsive design work described in section 4.2.1, since the more detailed robot illustrations became difficult to visually distinguish from one another at the smaller sizes needed to support a wider range of screen sizes. Badges were also reorganized into the two categories described above, which were not present in the first iteration. Neither the badge redesign nor the categorization was driven by a specific testing finding, since no participant in testing showed any interest in the gamification elements either way.

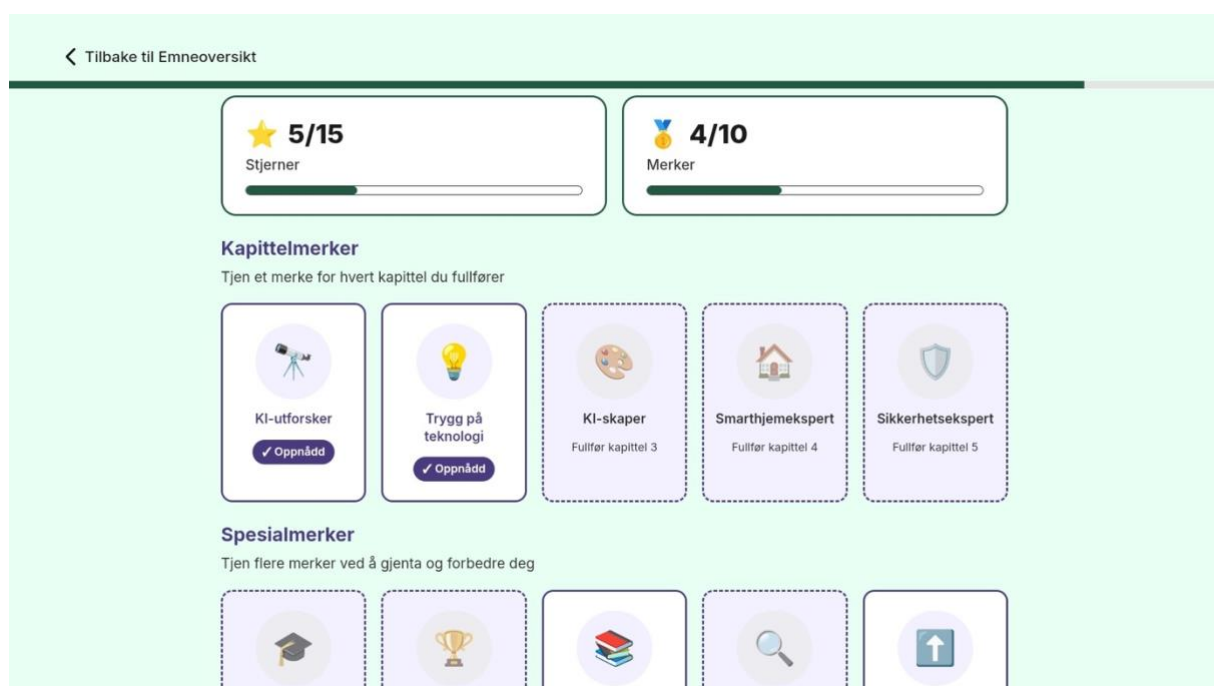


Figure 4.15: The progress overview page, lower portion, showing star and badge totals, and the badge collection.

## 4.2 Iterative design and development

### 4.2.1 Iteration 1: Figma prototype

The first iteration was tested with five participants (P1-P5, aged 62-82) using the Figma prototype on the tablet I provided. Several patterns emerged across the sessions.

## Navigation and scrolling

Scrolling itself was well understood, as all five participants scrolled through pages without difficulty, but every participant who scrolled down a page subsequently forgot that content existed above the fold, needing a prompt to check the top of the page again to find the progress button or the back button. All five participants also tapped directly on the back arrow-icon rather than the accompanying text label when navigating backwards, suggesting a habit carried over from phone use.

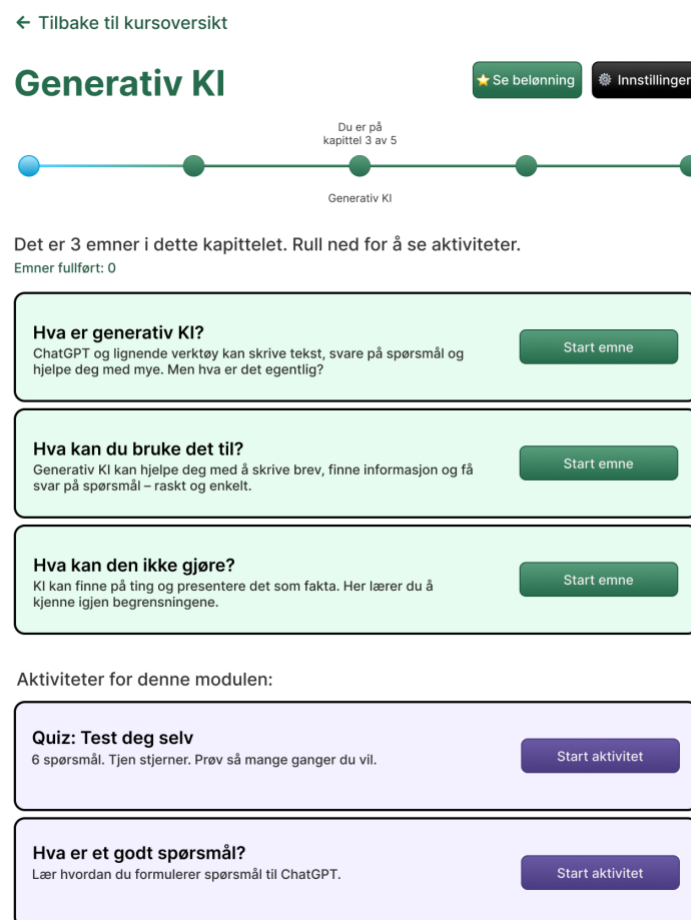


Figure 4.16: The Iteration 1 topic overview layout. Only two cards would be visible without scrolling on a tablet in landscape orientation.

## Wording and conceptual understanding

Separately, all five participants confused the topic and activity sections of a chapter, expecting the topics themselves to be the interactive activities when asked to describe

which activities a module had. Two of the five participants reported being unsure about the platform's underlying concept when asked directly at the end of the session. P2 suggested changing the careful wording of the quiz's wrong answer feedback from "Not quite" to a more direct "Wrong", citing that "We are adults. We can handle it.". P1 was confused with the wording "Scroll down to see activities" because the finger movement to perform scrolling down is upward. P2 also mentioned this and suggested to change the description to "See activities under the topics".

### Layout

The progress bar on chapter cards was not noticed by any participant, despite several reading the accompanying "6 of 6 topics completed"-text below it, suggesting it functioned as visual clutter rather than useful information.

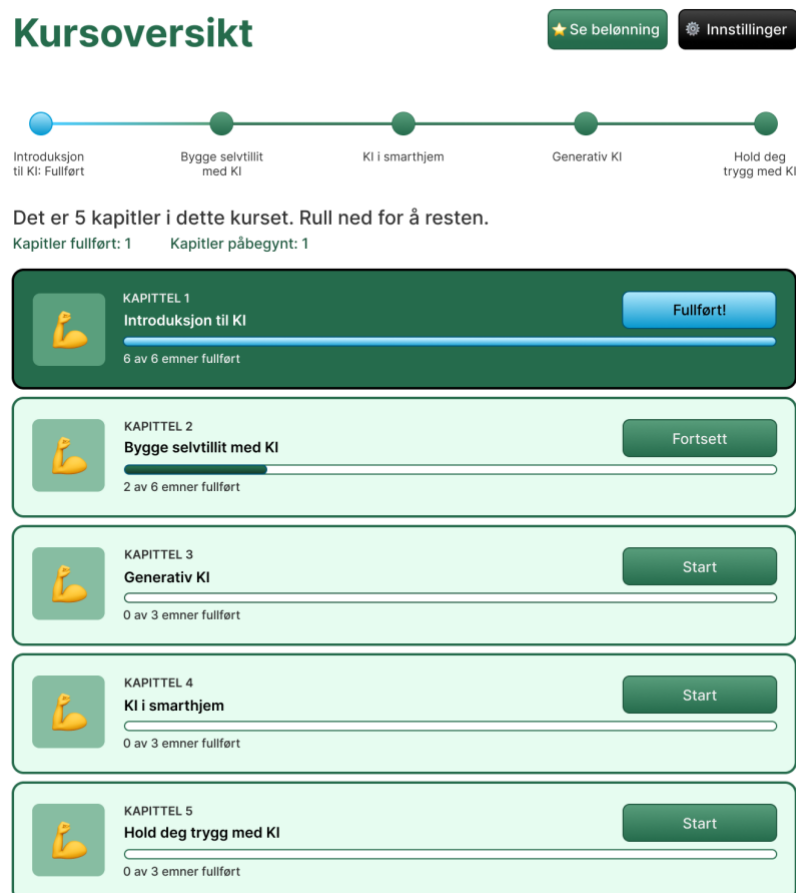


Figure 4.17: The Iteration 1 module overview layout, showing the progress bar on each topic's card.



### *Progress and rewards*

When asked specifically about which badges they had earned, most participants looked at the total count shown in a summary box rather than the individual badges themselves.

### *Quiz interaction*

All five participants initially pressed an answer option and paused, expecting this alone to register their answer, before independently discovering that a separate “Check answer”-button needed to be pressed. One participant was a little confused by the layout changing to show feedback after answering.

### *Gamification*

None of the five participants showed strong interest in the star and badge system during the session. This was treated cautiously rather than as a design failure, since participants had not yet engaged with the platform’s actual content and had not yet “earned” anything meaningful yet, and the gamification elements are generally expected to matter more once used over a longer period of time. One participant compared the badge system to Stolpejakt, a widely recognized Norwegian orienteering-style activity. This suggested that this comparison could be used to make the gamification concept more familiar to the target group, so it was used in my presentation of the project to participants in the next iteration of user tests.

### *Design changes made in response to iteration 1 testing*

- An onboarding flow explaining the platform’s concept, navigation, and functionality was added between the welcome page and the module overview page, directly addressing the concept confusion reported by two participants.
- A text size adjustment button was added to all pages.
- The button on top of the screen was changed to remain fixed at the top of the screen during and after scrolling, so they are always visible. This applies to the back, text size, progress, and info buttons.

- The module bullet-list stepper overview was moved from the module- and topic overview pages to the dedicated progress page, and replaced with a brief text summary of progress on the module overview. The bullet-list overview was also redesigned to a list of progress bars indicating completion of modules.
- The quiz feedback wording was changed to “Wrong”, and the layout was kept the same after confirming answers, with color and text showing feedback.
- The welcome screen was simplified to a single button, since the “continue where I left off” option was only relevant after a first visit, and the welcome page will only be visible the first time a user visits the site.

#### *Feedback noted but not acted on*

- Indifference towards gamification elements was not treated as a design problem, since the testing context did not give participants a chance to earn rewards meaningfully. This was left to be reassessed after the final round of testing, which would give the participants time to explore the platform and earn the rewards.
- The topic vs activity confusion was not resolved through a structural redesign at this stage, but was treated as a reminder of the importance of precise wording when labeling interface elements.
- The quiz’s separate “Check answer” button was deliberately kept, despite every participant expecting slightly different interaction, since it allows users to reconsider before committing to an answer. This is consistent with usability heuristics on preventing and easily correcting errors (Nielsen, 1994).

#### 4.2.2 Iteration 2: Coded prototype

The second iteration was tested with 3 participants (P6-P8, ages 60-83) using the coded React prototype. Several changes were made in response to Iteration 1 findings before this round of testing began. An onboarding page was added, the back button and “See progress” button was made to stay fixed at the top of the screen while scrolling, a text size control was added to the top of most pages, quiz feedback was changed from “Ikke helt” to “Feil”, and the module overview page was restructured into a two-column grid, removing the need to scroll to see all chapters. This round tested these changes directly and made new findings.

### *Onboarding*

The new onboarding page was read carefully by all three participants. The oldest participant (P8) assumed the onboarding content had ended after reading what was initially visible on screen, and needed a prompt to remind them it was possible to scroll down. This indicated the need for ways to show that the visible content may not **med** the only content available on a page.

### *Fixed header buttons*

Retesting the fix introduced after Iteration 1 showed that all three participants were able to see and use the button to reach the progress page via the fixed top bar. This was a clear improvement over Iteration 1, where all five participants needed a prompt to scroll up to locate the button. However, returning from that page back to the module overview was less consistent: one participant used the platform's back button successfully, one used the browser's own back button, and the oldest participant did not find a way back at all without help.

### *Topics versus activities*

The confusion about the distinction between topics and activities, which was found in all five participants of Iteration 1 testing, was only partially resolved. When asked to find a module's activities, one participant worked it out through reasoning ("*no, the quiz is probably the activity*"), but the other two showed the same confusion as before. Notably, I observed directly that all three participants had read the onboarding text explaining the topic/activity distinction, yet they still confused the two when it came time to locate them. This indicated that there was need for a more direct visual cue in addition to the introductory explanation.

### *Text size*

All three participants located and used the text size control immediately when asked, with no hesitation or need for hints, which can be considered a successful outcome for this new addition, which was added in response to an Iteration 1 participant's request for larger text.

## *Quiz*

The same hesitation after pressing an answer option as in Iteration 1 testing was encountered with all three participants here as well. Similarly to Iteration 1, all participants understood after a while that the answer must be confirmed. The oldest participant assumed the quiz was finished after answering the first question, and needed a hint that pressing again would move to the next question.

## *Recurring pattern with the oldest participant*

Across the session, the oldest participant repeatedly showed a pattern I noted as a form of hesitation bordering on giving up, rather than continuing to explore the page or read information text when something was unclear. This mirrors a similar pattern among the oldest and least digitally experienced participants in Iteration 1, suggesting this may be a more common characteristic of this specific segment of the target group, rather than something that can be addressed by any single design fix.

## *Design changes in response to Iteration 2 testing*

- Explanatory heading text above the topic and activity sections, as well as numbering the topics on the topic overview page, to make the intended starting point clearer for less experienced users and avoid the topic versus activity confusion.
- A visual scroll-position indicator and making next-buttons inactive until the user has scrolled to the bottom of the page, to imply that there is content further down, so users are less likely to miss important content.

## *Feedback noted but not acted on*

- The quiz's confirm before proceeding flow was reconsidered a second time, given it encountered the same hesitation among participants as test 1, but was again kept unchanged, for the same reason as before: it allow the user to reconsider and correct a mistake before committing to an answer, consistent with usability heuristics on easily reversible actions (Nielsen, 1994).
- As in Iteration 1, none of the three participants showed much interest in the gamification system. This was again treated cautiously, rather than as a design

failure, since the participants had not yet used the platform over a meaningful period of time and engagement.

#### 4.2.3 Design changes summary table

The table below summarizes the design changes made across both formative iterations, the usability finding that prompted each one, and where each change was subsequently tested.

| # | Originated in | Issue found                                                                                               | Change made                                                                                                                                | Tested in iteration # |
|---|---------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1 | Iteration 1   | 3 of 5 participants tapped through the two-button welcome screen without noticing a second option existed | Simplified to a single button; the second option was only relevant on return visits                                                        | 2                     |
| 2 | Iteration 1   | 2 of 5 participants reported being unsure of the platform's underlying concept                            | Onboarding page added, explaining course structure and content sourcing                                                                    | 2                     |
| 3 | Iteration 1   | All 5 participants forgot content above the fold after scrolling (e.g. rewards, back button)              | Back, rewards, and settings controls made to stay fixed at the top of the screen while scrolling                                           | 2                     |
| 4 | Iteration 1   | Per-chapter progress bar not noticed by any participant; treated as visual clutter                        | Removed from chapter cards; replaced with a brief text summary on the course overview, with full progress detail moved to a dedicated page | 2                     |
| 5 | Iteration 1   | Removing the progress bar freed space within each chapter card                                            | Course overview restructured into a two-column grid, removing the need to scroll to see all chapters                                       | 2                     |
| 6 | Iteration 1   | Text-size control introduced as the only remaining "setting," making a                                    | Settings button replaced with a persistent Help button, freeing the                                                                        | 2                     |

| #  | Originated in    | Issue found                                                                                                        | Change made                                                                                                             | Tested in iteration # |
|----|------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                  | separate settings page unnecessary                                                                                 | slot for always-available onboarding access                                                                             |                       |
| 7  | Iteration 1      | One participant requested larger text                                                                              | Text-size control added, adjustable from any page                                                                       | 2                     |
| 8  | Iteration 1      | Quiz feedback wording “Ikke helt” (Not quite) suggested as unclear by one participant                              | Changed to “Feil” (Wrong)                                                                                               | 2                     |
| 9  | Iteration 1      | All 5 participants confused chapter topics with activities                                                         | No structural change made yet; informed later design decisions                                                          | 2                     |
| 10 | Iteration 2      | Despite reading the onboarding text explaining topics and activities, participants still confused the two          | Explanatory heading text added above each section (“read topics in order” / “take the quiz after”), and topics numbered | 3                     |
| 11 | Iteration 2      | Risk that users could advance past a topic via the always-available “Next” button without reading its full content | “Next topic” button made inactive until the page is scrolled to the end, paired with a visual scroll-position indicator | 3                     |
| 12 | Iterations 1 & 2 | Quiz’s confirm-before-proceeding step questioned or reported as slow by some participants in both rounds           | Deliberately kept unchanged both times, to preserve the ability to reconsider an answer before committing               |                       |
| 13 | Iterations 1 & 2 | No participant in either round showed interest in the star/badge system                                            | Not acted on; treated as inconclusive given the short, first-time nature of the testing sessions                        |                       |

Table 4.1: Summary of design changes made across iterations 1 and 2, showing the finding that prompted each change and where it was subsequently tested.

A small number of additional visual changes, like simplifying the badge icons and splitting the badges into module and special categories, were made between Iteration 2 and the final deployed version. These were personal design decisions rather than responses to specific testing discoveries, and are therefore not included in the table above.

## 4.3 Final platform design

### 4.3.1 Accessibility

The platform was developed with a target of WCAG 2.1 AA compliance, in line with Norway's regulation on universal design of ICT solutions (see section 2.3.1). Several accessibility considerations were addressed directly through the design and testing process described in earlier sections of this chapter, and are summarized here alongside the results of dedicated accessibility testing.

#### *Text size and visual clutter*

Body text on the platform is set to 16px by default. This falls within general web conventions but below what a systematic review of font size for older adults specifically recommends for reading tasks on tablet devices: 18 points, based on a study of iPad users aged 62–84 (Chatrangsarn & Petrie, 2019, as cited in Hou et al., 2022), within a broader range of 14–22 points reported across the studies that review examined. Hou et al. (2022) also note that the preference for larger font sizes among older adults holds consistently across studies, regardless of device resolution, device type, or font typeface. However, a persistent text size control (available on most pages) allows a user to adjust text size up or down to a comfortable level at any time (see section 4.1.1). This adjustability, rather than a larger default text size, was the main mechanism to accommodate the target group's text size needs. Page layouts were also designed to minimize visual clutter. The module overview and topic overview pages, which present the highest density of information on the platform, were specifically restructured across iterations so that all content is visible without scrolling (see Table 4.1, item 5), reducing the need to hold information in memory while exploring the content of the page.

### *Not relying on color alone*

Consistent with WCAG Success Criterion 1.4.1 (Use of Color), information is never conveyed through color alone anywhere on the platform (W3C, 2025). Quiz feedback pairs color with text labels rather than relying on red and green alone, specifically to remain legible to colorblind users. Similarly, the distinction between a module's topics and its quiz uses three redundant cues together: position, color, and text.

### *Touch target size*

Given the target group's likely reduced fine motor precision, interactive elements were deliberately designed with generous touch targets. The text size control, for example, is split down the middle into two large tappable zones, rather than relying on the smaller minus and plus icons alone. Back navigation buttons are tappable across their full width, giving users a larger than usual tappable surface.

### *Semantic structure and screen reader support*

The platform uses semantic HTML and ARIA labels. Automated testing across six representative pages (welcome, module overview, topic overview, topic content, quiz, and progress overview) found a single <h1> per page with properly nested subheadings, defined landmark regions (header, main content, navigation, footer), and descriptive aria-label attributes on every icon control. Interactive quiz elements use appropriate semantic roles.

### *Automated accessibility testing*

The WAVE web accessibility evaluation tool (WebAIM, 2026) was used to scan the six pages. Results are summarized in Table 4.1. No structural or coding errors were found on any page tested. All flagged contrast errors were isolated to the course overview page, discussed further below.



| Page             | Errors | Contrast Errors                    | Alerts | AIM Score  |
|------------------|--------|------------------------------------|--------|------------|
| Welcome          | 0      | 0                                  | 0      | 10/10      |
| Course overview  | 0      | 2–3 (varied across repeated scans) | 0      | 8.2–9.1/10 |
| Chapter overview | 0      | 0                                  | 0      | 10/10      |
| Topic content    | 0      | 0                                  | 0      | 10/10      |
| Quiz             | 0      | 0                                  | 0      | 10/10      |
| Progress         | 0      | 0                                  | 0      | 10/10      |

Table 4.2: Automated accessibility results

### Manual keyboard navigation

Tab order was checked manually by navigating the platform using only the keyboard, which worked as expected throughout. This was cross-checked against WAVE’s navigation order analysis, which confirmed a logical order matching the visual layout. Testing with dedicated assistive technology such as a screen reader was not conducted, and is noted as a limitation in section 5.2.8.

### Color contrast

Representative text and interface color pairs were checked using the WebAIM Contrast Checker (WebAIM, n.d.) including body text, headings, and quiz interface colors. Where an element used partial opacity rather than a flat color, such as the secondary status text on completed module cards, which are rendered in code as completely white at 85% opacity, the effective color was sampled directly from the page using a dropper tool, rather than checking the source value alone. This particular pairing (#E0E7E4 against the dark green #215A41 card background) measured 6.42:1, passing WCAG AA for normal text (4.5:1 required) though not AAA (7:1 required)(W3C, 2026), which is a reasonable outcome given AA, not AAA was the project’s compliance target. All other checked pairs, including body text, headings, and quiz interface colors, passed WCAG AA and AAA for both normal and large text, with ratios ranging from 6.42:1 to 16.6:1, comfortably above the required 4.5:1 and 3:1 minimums.

The one exception found by automated scanning involves the star icon used to indicate quiz performance on completed module cards. The icon is rendered as a system/platform emoji glyph rather than a custom drawn icon, and is made up of internal color tones: light gold highlights and a darker gold shadow. WAVE scans of the page flagged instances of very low contrast, which is noted as a black against dark green pairing, though the black is not visible to the naked eye even when the page is substantially enlarged. Manual checking found that the darkest gold color from the star icon fails the 3:1 minimum requirement while the lightest gold passes comfortably. This points to either the star icons having an almost invisible black outline failing the check, or that it is the star icons' darkest color that fails the contrast check.

### 4.3.2 Visual design

The platform's visual language was established during the Figma design stage and carried through into the coded implementation, using CSS custom properties to keep colors, typography, and spacing consistent and easy to update across the codebase.

#### *Color system*

The platform uses a two-color system: a primary dark green for its main content and topics, and a secondary dark purple (#4A3A80) for quizzes, activities, and badges. Each of these colors are paired with a lighter variant for gradients on buttons, and a very light tint for backgrounds. Neutral tones like white and light gray, and the red feedback colors round out the palette.

| Token                 | Hex     | Swatch                                                                              | Use                                          |
|-----------------------|---------|-------------------------------------------------------------------------------------|----------------------------------------------|
| --color-primary       | #215A41 |  | Primary green — main content, topics         |
| --color-primary-light | #5A9F7D |  | Primary green, lighter variant for gradients |

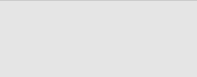
| Token                | Hex     | Swatch                                                                              | Use                                             |
|----------------------|---------|-------------------------------------------------------------------------------------|-------------------------------------------------|
| --color-purple       | #4A3A80 |    | Secondary purple — quizzes, activities, badges  |
| --color-purple-light | #6B5BA6 |    | Secondary purple, lighter variant for gradients |
| --color-bg-green     | #E8FFF4 |    | Background tint, green content                  |
| --color-bg-purple    | #F3F0FF |    | Background tint, purple content                 |
| --color-text         | #1A1A1A |    | Primary text                                    |
| --color-text-muted   | #2E2E2E |    | Muted / secondary text                          |
| --color-white        | #FFFFFF |    | Neutral                                         |
| --color-border       | #E5E5E5 |   | Neutral, dividers/borders                       |
| --color-error        | #C62828 |  | Feedback — errors                               |
| --color-bg-error     | #FDECEA |  | Feedback — error backgrounds                    |

Table 4.3: List of colors used in the platform

The consistent mapping of green to topics and purple to activities also supports faster recognition. Once a user has learned the association, the color functions as one of several redundant cues (alongside position and text labeling, see section 4.3.1) that lets them identify what kind of screen they are on without having to read and interpret the page content each time. This consistency is also deliberately limited to a small, functional palette rather than a wide color-coding scheme, to avoid adding visual clutter: Kaur and Chen (2022) found crowded interfaces in 3 of the 10 apps they evaluated for older adults.

### Typography

Body text uses inter, with a system-ui/sans serif fallback. Font sizes are defined in rem rather than fixed pixels throughout the platform. Body text sits at 1 rem (16px), h1 page titles scale from 2 rem (32px) on mobile up to 2.5 rem on tablet portrait and 2.75 rem on tablet

landscape. UI labels and meta text sit at 0.9375 rem (15px) and supplementary text, such as source attributions, ranges from 0.8125 to 0.875 rem (13-14px). Every size is wrapped in *calc(X rem \* var(--font-scale, 1))*, so the text-size control described in section 4.3.1 scales all text on the platform proportionally through a single shared multiplier. The control offers six discrete steps, from 0.875x up to 1.5x, with 1x being the default, meaning body text can range from 14px at the smallest step up to 24px at the largest, and the tablet landscape h1 title, at 44px default, can reach 66px at the maximum step.

### *Reusable components*

Wherever possible, elements were defined once and reused everywhere, built as a small set of shared components. TopicCard is the clearest example. One component renders both a topic and a quiz, switching between the green/purple mapping discussed above through a single variant prop. ModuleCard works the same way for module cards, switching between an in-progress and a completed state. There is no separate “not started” state. A module with zero progress renders identically to one that has been partially completed, distinguished only by the completion count in the card’s text.

### *Visual tone*

The overall visual tone was kept deliberately simple. A largely single-color palette, with purple reserved specifically for activities, rather than a wider range of colors across the interface. This followed the same clutter avoidance reasoning already discussed in the Color System section, where the goal was to avoid distracting users from the most important elements on each page, whether through a crowded layout or through excessive use of color. This was a personal design decision rather than one driven by usability testing findings, though it aligns with Nielsen’s (1994) aesthetic and minimalist design heuristic, which cautions against interface elements that compete with more relevant information for a user’s attention.

### 4.3.3 Gamification elements

The platform incorporates a simple set of gamification elements intended to support motivation and a sense of progress without introducing stress or pressure. Gellner and Buchem's (2022) recommendation against quizzes (see section 2.4) is worth addressing here, since quizzes are an important part of SeniorKI. Their finding evaluated quizzes specifically as a motivational technique, distinct from a comprehension check tool tied to specific content. In my platform, quizzes serve the function of assessing topic comprehension, rather than as an optional engagement device. The star and badge rewards are the gamification elements, and the quiz itself is pedagogically independent of its motivational framing.

The gamification elements are:

#### Stars

Each quiz awards between zero and three stars based on the number of correct answers. Stars are displayed at the end of the lesson as a summary, along with an encouraging message, giving the users an immediate sense of how well they performed.

#### Badges

The platform contains a possible ten badges in total, split between two categories. Five module badges are tied to completing a module, while five special badges reward continued use beyond first completion, like repeating a module or repeating all modules. Badges are displayed on the Progress page that can be entered from most other pages via the "Se fremgang" (See progress) button.

#### Progress overview

The Progress page overview serves both as information and as motivation, where users can see at a glance how far they have come and how much remains. It provides the user with a visual representation of their progress through the course through a list of progress bars. These bars display the user's progress in each module, as well as the module's status of

either completed, in progress (with a completion count), or not started, with icon and text label for each.

## 4.4 Curriculum structure

The curriculum is based directly on the five-module structure proposed by Kaur et al. (2025), adapted for a self-paced online format and a Norwegian audience. All user-facing content is written in Norwegian. The final chapter order deviates slightly from the sequence proposed by Kaur et al. (2025), as the order of modules 3 and 4 is swapped, moving Generative AI ahead of AI in Smart homes. This was an intuitive decision rather than a choice grounded in a specific source. Because of how frequently generative AI tools such as ChatGPT are discussed in Norwegian media, I expected this topic to be more immediately engaging to participants than smart home applications, and moved it earlier in the sequence to keep participants' interest and attention before introducing a topic they might have less prior familiarity with.

Kaur et al.'s (2025) curriculum also specifies a practical activity for each module: exploring common AI applications (Module 1), a discussion prompt about privacy and security experiences (Module 2), hands-on practice using ChatGPT for writing assistance (Module 3), a roleplay exercise asking AI for help with daily tasks (Module 4), and a case study on spotting AI-related misinformation online (Chapter 5). Due to time constraints, these activities were not implemented in the platform. Each module was scoped down to written content and a quiz only. This is discussed further as future work in section 5.3.1. The full content of the curriculum is available in Appendix H.

Before being added to the platform, all module content underwent a language level review by an AI large-language model (Claude), given a detailed prompt describing the target group and intended tone. This was a fallback for an originally planned review by Seniornett course leaders, which did not take place due to time constraints (see section 3.2 for the full review process and its limitations).

## Module 1: Introduction to AI (Introduksjon til KI)

This module provides an introduction to what AI is, including a basic explanation of how it works (simple descriptions of machine learning and neural networks with relatable examples), and where it shows up in everyday life (personalized recommendations on Netflix, facial recognition to unlock a phone). It also addresses common myths and misconceptions, and hallucinations.

## Module 2: Building confidence with AI (Bygge selvtillitt med KI)

This module addresses fears and concerns older adults have about AI, and covers privacy and security, including that personal data entered into AI tools may be stored and used to train future versions of the tool. It also covers how AI learns and makes decisions, including the difficulty of knowing how an AI system produces a specific answer, often called the “black box”-problem, and how AI systems can reflect biases from data they were trained on.

## Module 3: Generative AI (Generativ KI)

This module introduces generative AI tools like ChatGPT and covers what they can and cannot do, as well as how to use them for writing assistance and hobby activities by writing better prompts.

## Module 4: AI in smart homes (KI I smarthjem)

Focuses on AI technologies that older adults are increasingly likely to encounter in domestic settings, including voice assistants, smart doorbells, robot vacuum cleaners, and other AI-powered home devices.

## Module 5: Staying safe with AI (Hold deg trygg med KI)

Addresses safety in an AI-enabled world. Covers how to identify trustworthy AI sources, common AI-related scams and precautions one may take to avoid them, and to understand the ethical dimensions of AI use.

Each module has a 3-question quiz that awards the user with zero to three stars, one per correct answer, and a badge upon completion. There are also badges available for completing modules a second time, to encourage repetition and support knowledge retention.

## 4.5 Summative evaluation

### 4.5.1 Quantitative evaluation results

Of the seven participants who completed both the pre- and post-test, five improved their knowledge test score after using the platform, ranging from a 1-point to a 5-point gain, and two showed no change, having already scored the maximum on the pre-test. A Wilcoxon signed-rank test was used to test whether this improvement was statistically reliable. Since the two unchanged pairs carry no information about direction of change, they were excluded from the test, which left an effective sample of five pairs. As the research question asks specifically whether the platform improves understanding, rather than whether scores change in either direction, a one-sided test was applied. The result was statistically significant,  $p = .031$ , which indicates that the observed improvement is unlikely to be due to chance alone. For transparency, the two-sided result is also reported:  $p = .063$ , which does not reach the conventional significance threshold. The difference reflects the small effective sample size rather than a weaker underlying pattern. All five participants who had room to improve did so, which is the most consistent result the test can register at this sample size. Quantitative evaluation results can be seen in Appendix I.

Time spent between the pre- and post-test varied widely, from 15 minutes to just under 49 hours. Three participants completed the entire process, including the pre-test, platform content, and post-test, in under an hour: P11 (15 minutes), P13 (39 minutes), and P6 (55 minutes). The remaining four took between roughly 7 and 49 hours. No participant used the platform for more than two days.



All seven participants who completed the post-test scored the maximum of 10 out of 10, meaning every improved score reached the same ceiling value. This pattern is discussed as a limitation in section 5.2.3, alongside a related concern regarding the knowledge test's answer distribution.

Participants rated the platform's usability using the System Usability Scale (SUS). The average score was 89.3. On Bangor, Kortum, and Miller's (2008) adjective rating scale, scores above 87.5 fall within their "Best imaginable" category, placing the platform at the upper end of their scale. On their alternative "university grade" analog, where a score of 90 or above is an A, 89.3 falls just short, and scores a B, bordering on their highest "Best imaginable" category. Following Lewis and Sauro's (2009) division of SUS into two subscales, the platform scored slightly higher on Learnability (96.4) than on the broader Usability subscale (87.5), suggesting that participants found the platform particularly easy to get started with without outside help.

A closer look at the individual item responses shows two patterns worth reporting. Three participants, P6, P7, and P14, gave identical responses across all ten SUS items (5-1-5-1-5-1-5-1-5-1), each corresponding to the maximum possible score of 100. P2's responses matched the expected response pattern on 8 of the 10 items, but included the lowest possible rating on two positively worded items, S3 ("I thought the system was easy to use") and S9 ("I felt very confident using the system"). This pattern does not match the polarity of P2's other responses or their positive open-text comments elsewhere in their post-test (see section 4.5.2), and is plausibly explained by a misreading or misclick on these two items rather than a genuine negative assessment.

#### 4.5.2 Qualitative evaluation results

In addition to the SUS-questionnaire, participants were asked two open-ended questions about their experience with the platform, plus an additional comment on the star and badge gamification elements. Response rates varied by question: 6 of 7 participants answered whether anything was difficult to understand, 5 of 7 answered whether anything was lacking, and 4 of 7 left an optional comment on the gamification elements.

Most responses were short and positive, with several participants reporting no notable difficulty (P2, P6, P11) and nothing was lacking from the course (P6, P7, P11, P13).

Two responses offered more specific reflection. P10 noted that they have a background in ICT and that the platform was easy for them to understand, but suggested it might be more difficult for people with less interest in ICT. P14 described being fooled by similarly worded answer options on the quizzes: *"Some of the answer options were a bit tricky. I fell into the trap because I didn't read the question carefully enough"* (translated from Norwegian).

A more consistent pattern appeared in the optional comments on the gamification elements. Although both P10 and P14 rated the system positively on the accompanying 1-5 scale (4 out of 5, in both cases), both independently noted that immediate feedback on whether an answer was correct (on the quiz, presumably), mattered more to them than the gamification elements themselves. P10 wrote: *"It's an encouragement to keep going. I thought it was more important to get confirmation of right and wrong answers"* (translated). P14 made a similar point: *"The most important thing was feedback on right and wrong answers. But some people do enjoy medals and stars"* (translated).

Several participants also commented positively on the platform's instructional clarity and that it was appropriate for the target group. P13 described the instructions as containing "a lot of information", delivered clearly, and P10 characterized the course as built to give insight into what AI is "at a level suited to the age group".

## 5 DISCUSSION

This chapter discusses and interprets the results presented in Chapter 4, considering the project's research questions, its own data, and existing literature. Section 5.1 discusses specific findings from the formative and summative testing rounds, section 5.2 addresses the project's limitations, and section 5.3 outlines directions for future development.

Beyond its contribution to research, this project also has practical value. SeniorKI is a free, publicly available platform, not a prototype limited to this study, and it remains usable by Seniornett or others in the target population until at least June 2027, when the current domain registration expires. It is also, to my knowledge, the only working implementation of Kaur et al.'s (2025) curriculum, rather than a proposal for one. The design and evaluation choices made throughout this project may also be useful more broadly to others developing similar tools for older adults.

### 5.1 Results discussion

#### 5.1.1 Demographic information

Because Kaur and Chen's (2023) and Kaur et al.'s (2025) research shaped both the problem this thesis addresses and the curriculum it implements, it is worth comparing this project's sample to theirs, to assess how well these findings might apply to the population the project is meant to serve. The sample was equally balanced by gender: of the 12 unique participants across all three iterations, 6 were female and 6 were male, closely matching the near-even split reported by Kaur and Chen (2023). The age distribution, however, skewed younger than Kaur and Chen's sample: participants in this project ranged from 60 to 83 years, with half (50%) aged 60-69, a quarter (25%) aged 70 to 79, and a quarter (25%) aged 80 to 89. By comparison, of Kaur and Chen's participants, roughly 82% were 70 years or older.

Education level was more mixed in this project's sample, with a slight lean towards lower education: 3 participants had completed high school, 4 had a vocational qualification, and 5 held a college or university degree (2 with 1-3 years, 3 with 4 or more years). This differs

from the sample in Kaur et al. (2025), in which 68.6% of survey respondents held at least a college degree. Only 5 of 12 (42%) in this project had a comparable level of education.

Kaur et al. (2025, p. 222) note this as a limitation of their own study: because of how participants were recruited, roughly two thirds had above average education and may already be *“better equipped and more motivated to learn and use AI-enabled products and services than the broader older population”*, which they suggest that may limit how well their findings reflect the broader population’s needs. This project’s sample is more educationally diverse, so it is not subject to the same limitation. If anything, it might help explain a pattern noted in the next section of this chapter (section 5.1.2): the action paralysis observed in the oldest, least experienced participants may partly show a mismatch between the curriculum’s foundations, developed with a more digitally skilled and already interested population in mind, and the needs of users who are neither as skilled nor as motivated to begin with.

### 5.1.2 Action paralysis among the least experienced participants

A pattern noted independently in both formative testing rounds was that the oldest, least digitally experienced participants in each group tended to respond to confusion by disengaging rather than exploring further. Rather than reading available information text or trying another part of the screen, these participants tended to stop and wait for help, or assumed they had reached the end of what they could do. This happened with 3 different participants across Iteration 1 and Iteration 2, which suggests that it reflects a real pattern within this segment of the target group, rather than something specific to one individual.

This pattern is harder to design around than most usability findings from the testing rounds. The usual response to confusion would be to add more explanatory text, but the participants affected by this pattern were also the ones least likely to read the text in the first place. Adding more of the same kind of help was therefore unlikely to reach the users who needed it the most. No design change addressing this pattern directly was identified during the project, and it remains an unsolved issue.

One possible explanation is a lower sense of mastery over digital tools among the oldest and least experienced participants. A national Norwegian survey found that self-reported mastery of internet use declines sharply with age: 80% of 61-70-year-olds report feeling they handle internet use well, compared to 64% of 71-80-year-olds and 43% of those aged 81-100 (Slettemeås & Mainsah, 2025). If a participant already expects to struggle, giving up at the first sign of confusion may feel like the safer response compared to continuing to explore and risk making further mistakes. This explanation fits the observed pattern, but it was not tested directly in this project and should be treated as a plausible interpretation rather than a confirmed one.

### 5.1.3 P7/P13 knowledge gain

P7 shows the largest knowledge test improvement in the sample, from 5 to 10. For the sake of transparency, this result deserves closer attention, since a plausible alternative explanation exists aside from genuine learning. P13, a family member in semi-regular contact with P7, completed both the pre- and post-test on June 17<sup>th</sup>. P7 did not begin the test until the following day, June 18<sup>th</sup>. Since both participants took the same, identical test, it is possible that P13 shared some of the test content with P7 beforehand, whether intentionally or in passing conversation, which could explain part of, or all of P7's improvement. In my own assessment, this seems unlikely given the nature of the relationship between these participants, and course content is not something I would expect to come up between them. However, this is a subjective judgement, and not something the available data can confirm or rule out, and genuine learning and prior exposure to the test content should both be treated as plausible explanations for this result.

To check how much this one data point affects the main quantitative finding, the Wilcoxon signed-rank test reported in section 4.5 was recalculated with P7 excluded. With P7 included, the effective sample of five participants showed a statistically significant improvement using a one-sided test ( $p = .031$ , see section 4.5). With P7 excluded, leaving four participants (P6, P10, P13, and P14, all with positive gains between 1 and 3 points), this significance no longer holds ( $p = .063$ ). This means the significance of the main quantitative result depends partly on the one data point that could be criticized due to possible prior

exposure to the test content. This does not mean the underlying improvement is not real, since four of four remaining participants still improved, but it does mean the result is more fragile than the original reported p-value alone suggests.

#### 5.1.4 Completion times and self-paced use

None of the seven summative completers used the platform for more than two days, and three completed both the pre-test, the platform content, and the post-test in under an hour. P11 finished in 15 minutes, P13 in 39 minutes, and P6 in 55 minutes. The remaining four took considerably longer, ranging from just over 7 hours to just under 49 hours (see section 4.5.1). This pattern complicates two aspects of how the test results should be interpreted.

First, the summative evaluation gave participants a two-week period to use the platform at their own pace, rather than expecting it to be completed in a single sitting. None of the completers used it this way. This does not invalidate the knowledge test or SUS results, but it means the summative evaluation mainly captured a single, short first encounter with the platform, rather than the sustained use the test was intended to evaluate.

Second, this directly affects gamification and engagement. The platform does not track individual usage sessions, so the number of sessions cannot be confirmed directly, only inferred from the elapsed time between the delivery of the pre- and post-test. The three fastest completions (P6, P11, P13) most plausibly points to a single sitting. P2 and P14, whose elapsed times spanned roughly two days, and P7, whose 17-hour gap crossed overnight, more plausibly involved at least two or more sessions. Even so, none of these show evidence of sustained engagement across the full two week window the evaluation was meant to capture. The evaluation therefore cannot speak confidently to whether the platform's gamification elements encouraged the kind of continued use over time they were intended to support (see section 2.4). The star and badge system might still be effective at encouraging return visits, but this project's data offers only weak and partial evidence either way.

### 5.1.5 SUS responses

The identical and maximally polarized SUS responses from P6, P7, and P14 (see section 4.5) are worth interpreting cautiously rather than taken as three independent confirmations of an excellent experience. One possible explanation is extreme response style: a tendency to favor the endpoints of a rating scale over its midpoint. Research on extreme response style and age has produced mixed findings overall, but at least one large study found a significant age-related increase, partly explained by age-related decline in cognitive processing (Schneider, 2018). Completion times varied considerably between these three participants, however (55 minutes for P6, over 17 hours for P7, and just over two days for P14, see section 4.5.1), which complicates a single explanation covering all three. A fast, low effort completion is more consistent with a shortcut-heavy response style than a completion spread across multiple days. This pattern is noted as a limitation of interpreting the SUS scores at face value.

### 5.1.6 Gamification elements

The comments on stars and badges from the summative evaluation (see section 4.5) offer a point of contact with Gellner and Buchem's (2022) finding that quizzes were rated the least motivating gamification technique among older learners, while instant feedback was rated the most motivating. P10 and P14 independently prioritized quiz answer feedback over the reward system, a pattern that echoes this same contrast. Read alongside the functional distinction drawn in section 4.3.3, between quizzes as content assessment and quizzes as a motivational technique, this suggests that SeniorKI's quizzes might be experienced differently from those evaluated in Gellner and Buchem's study. This should be treated cautiously, as it rests on two open-text comments within an informal, single coder qualitative analysis (see section 3.4), and shows only that feedback was valued more than reward, not that the quiz format itself was found more tolerable. Whether SeniorKI's quizzes were experienced as tedious in the way Gellner and Buchem's participants reported remains untested by this project's data.

### 5.1.7 Reflections on AI-assisted development

Using Claude Code for most of the implementation (see section 3.2) came with a clear trade-off between speed and control. Design and functionality changes could be made much faster than manual coding would have allowed, making it possible to try larger changes without a large cost in development time. At the same time, having AI generate most of the code in large chunks made it harder to maintain the same overview of the codebase that manual coding usually gives. Because manual edits were rarely needed, there was also less of the incidental exposure to the code that normally comes from writing and reviewing it by hand, which made it easier for small details to go unnoticed.

One clear example of this came up late in development. Font sizes were set in rem units rather than fixed pixels, specifically to support the text-size control described in section 4.3.2. It was only later that I noticed this had been applied more broadly than I asked for. Almost all the sizing on the platform, including spacing, margins, and padding, also used rem. This is not a problem in itself, and it likely makes the design more consistent overall, but it shows the trade-off clearly. Rem values matching specific pixel sizes are often awkward numbers to work with (0.8125rem for 13px, for example), which makes them harder to adjust manually than a pixel value would be. If I had written this code by hand, I would have noticed this pattern immediately. With AI-assisted development, it went unnoticed until a later, unrelated edit needed to be made.

This does not undermine the value of this approach. The speed gain was real, and it made iterations possible that likely would not have fit within the projects timeframe otherwise. It does suggest, however, that this kind of AI-assisted development carries a risk of small, unnoticed patterns creeping in, whether helpful or not, and that taking time to review AI-generated code is a worthwhile practice for catching this kind of thing earlier.



## 5.2 Limitations

### 5.2.1 Bias

All seven participants who completed the summative evaluation had a pre-existing personal relationship with me in some form prior to the study: three through family ties and four through repeated social contact built up during my ten weeks of volunteer work at Seniornett datacafés. No participant in the final sample was recruited without some form of prior contact with me. Three of the seven had also had prior exposure to the platform itself from participating in an earlier iteration.

This raises two concerns: Firstly, the participants with a personal or family connection to me may be more inclined to respond favorably out of politeness or a wish to support the project. This would primarily affect self-report measures (the SUS questionnaire and open-text responses) rather than the knowledge test, which has objectively correct answers. Secondly, the initial recruitment pool was broader than the final group of participants that finished their participation, and included several Seniornett members with little to no contact with me. These participants appear to be overrepresented among those who signed up but did not complete the study, compared to participants with an existing relationship to me.

This pattern suggests that weaker ties to me may correspond to lower follow-through, though no systematic data on reasons for non-completion was collected. Seen together, these two effects mean that the final sample may be biased toward participants that are both more positively disposed towards me and more committed to completing the study, which limits how far the results, particularly the self-report measures, can be generalized to older adults without such a relationship towards me.

### 5.2.2 Small sample sizes

Several parts of this project were affected by smaller sample sizes than originally planned or recommended. In the second formative usability test, one of four recruited participants cancelled at the last minute, reducing the group to well below Nielsen's (2000) recommended number of five per iteration (see section 3.3.2). Since Nielsen's own estimate is that five participants surface around 85% of usability problems in an interface, testing with only three in this round means it is likely that not all usability problems in the coded prototype were identified before the platform was evaluated more broadly in the summative test. This is partly offset by the iterative structure of the testing process itself: Nielsen (2000) also notes that several smaller tests conducted across iterations tend to uncover more issues collectively than a single larger test, and Iteration 2's findings build on Iteration 1's full five participant round. This does not fully compensate for the shortcomings of Iteration 2 entirely, since usability problems introduced by the design changes between iterations would only have had three participants' worth of opportunity to surface.

The summative evaluation's knowledge test result was also based on a small sample. Of the seven participants who completed both the pre- and post-test, two showed no difference and were excluded from the Wilcoxon signed-rank test. This left an effective sample of five (see section 4.5). This is already a small sample for any statistical test, and the sensitivity analysis in section 5.1.3 shows that excluding yet another participant, P7, whose result could be argued to have doubtful validity, reduces the effective sample to just four and removes the statistical significance of the result.

Across both the formative and the summative rounds, the project's conclusions rest on samples that are smaller than what is ideal. This does not mean the findings are unreliable, but it does mean they should be read as indicative rather than conclusive. Any single result, whether a usability finding from Iteration 2 or the Wilcoxon test's p-value, is better treated as one data point supporting a broader pattern than as a definitive, generalizable conclusion on its own.

### 5.2.3 Knowledge test ceiling and answer distribution

All seven participants who completed the post-test scored the maximum possible score of 10 out of 10, meaning the knowledge test could not distinguish between different degrees of **improvement** once a participant reached full marks. Five out of seven pairs improved by between 1 and 5 points to reach this ceiling, while two had already scored the maximum on the pre-test. This ceiling effect limits how the pre/post comparison can be interpreted: it shows that no participant's understanding measurably worsened and that most of them improved, but it cannot show whether some participants understood the material more thoroughly than others once all reached the same maximum score.

A related concern is that the correct answer to the knowledge test is option C in 7 of the 10 questions. A participant who systematically guessed C on questions they were unsure of could score as high as 7 out of 10 without demonstrating genuine understanding, and could reach the observed ceiling of 10 out of 10 with only a small number of confident, correct answers on the remaining questions. This does not necessarily mean the observed improvements reflect guessing rather than learning, but the possibility cannot be ruled out with the available data, and it is a design flaw in the test itself rather than a property of participants' behavior. Future iterations of the knowledge test should balance the distribution of correct answers across the response options to remove this possibility.

### 5.2.4 Attrition

Attrition occurred at two points in the recruitment and testing process. Of the 15 people who initially signed up for the summative evaluation, 9 went on to complete the pre-test, while the remaining 6 did not begin it. Three of these six explained their withdrawal by email (unprompted) around the time the study began. One cited a health issue unrelated to the course. Another cited a change in personal circumstance that left no time to participate. A third stated that they did not consider themselves skilled with computers and withdrew for that reason. The remaining three did not provide a reason.

Of the 9 participants who began the pre-test, 7 went on to complete the post-test as well. One of the two who did not (P12, a couple who completed the pre-test together) explained

by email that the course felt too extensive for people their age. The other, P9, did not provide a reason for discontinuing.

Taken together, the stated reasons for withdrawal are mixed: two relate to circumstances external to the course (health, personal circumstances), while two directly reference a lack of confidence or comfort with computers and digital tasks. This latter pattern is notable given that low digital confidence is a characteristic that is part of the platforms' intended target group. It suggests that some of the same difficulties the course aims to address may also be a barrier to initial participation, independent of the course's own design or content.

#### 5.2.5 Single researcher qualitative analysis

The qualitative analysis of observation notes and open-text feedback, across all three testing rounds, was conducted by a single researcher, without a second coder to independently verify the categories identified or to check for agreement between interpretations. As noted in section 3.4.1, this means the analysis is better described as informal, category-based grouping inspired by thematic analysis, rather than a full application of the method. As a sole coder, I also designed the study and collected the data myself, making me what Lazar et al. (2017) calls a subjective or inside coder: someone whose prior familiarity with the material can shape interpretation without their awareness. This is a recognized trade-off in qualitative research, and Lazar et al. (2017) note that single-coder analysis is generally considered acceptable in practice, provided the approach and its limitations are clearly reported, as done here.

#### 5.2.6 Effect of the test situation on observed behavior

Several participants in the formative usability tests appeared to experience some stress from the testing situation itself, despite my efforts to keep the sessions relaxed and to reassure participants that the platform, not their ability, was being tested (see section 1.3). This raises a validity concern of what the formative tests could observe. Participants who feel some pressure to perform may read screen content more quickly, or less thoroughly, than they would if using the platform alone and at their own pace. Several findings from the formative

rounds, such as participants not noticing the progress bar or not reading information on the screen, may partly reflect this time pressured, observed context, rather than reflecting exactly how a participant would engage with the same content unsupervised. This does not necessarily invalidate these findings, but it is a relevant factor to consider when interpreting whether the observed behavior reflects real world use.

### 5.2.7 Small font size

Base body text size (16px) is smaller than the 18px recommended in accessibility guidance for older adults (Hou et al., 2022), relying on the platform's adjustable text control to compensate rather than a larger default. In addition, subheadings within topic content are the same size as surrounding body text, distinguished only by font-weight rather than size. Neither issue was identified through any of the three rounds of user testing, and they are just noted here as opportunities for refinement rather than confirmed usability problems.

### 5.2.8 Accessibility testing methodology

Accessibility on the platform was primarily assessed using automated tools (WAVE) and manual checks conducted by myself, rather than through expert accessibility review or user testing involving people with disabilities (see section 4.3.1). This is a meaningful limitation. Automated accessibility scanners are effective at catching missing features like headings, alternative text, and sufficient color contrast that are defined by a set of guidelines such as WCAG, but are documented to miss context-dependent problems that require human judgement, such as whether alternative text is actually meaningful to a user (Mateus et al., 2021). In one case study of an e-commerce website, automated testing identified several WCAG violations, but manual expert review and testing with blind users uncovered additional problems, including confusing navigation and inadequate feedback, that the automated scan did not detect (Gonçalves et al., 2018). It is possible that SeniorKI has accessibility issues of this kind that neither the automated scans nor my own manual reviews were able to catch, particularly because no dedicated screen reader testing was conducted (see section 4.3.1).

## 5.3 Future work

### 5.3.1 Interactive creative activities

Kaur et al.'s (2025) curriculum specifies a practical activity for each module, including a roleplay exercise asking AI for help with daily tasks and hands-on practice writing with ChatGPT, but due to time constraints, SeniorKI implements only written content and a quiz for each module (see section 4.4). This is likely the platform's most significant design limitation, and therefore warrants future development. Gellner and Buchem (2022) found that empowerment of creativity and feedback, described as the ability to learn by being creative and trying things out, was rated the second most motivating factor among the eight core drives they tested, behind only seeing one's own progress. SeniorKI's quiz-based format does not give users a chance to try things out with AI directly. Understanding is tested, but not built through hands-on experimentation. A natural direction for future development would be interactive, exploratory activities that let users experiment directly with AI tools, such as trying prompts with a chatbot, rather than just reading about how such tools work.

### 5.3.2 Blended learning

Blended learning was deliberately left out of scope for this project (see section 1.2), and the platform remains a purely self-paced digital course. The findings from the evaluations, however, suggests that this is worth revisiting rather than completely ruling out. The action paralysis pattern observed in the oldest, least experienced participants (see section 5.1.2) is difficult to address through the platform's own design, since the participants affected by it were also the ones least likely to read additional information on the screen when confused. A brief, scheduled in-person session with a volunteer present, rather than more text on the screen, is a more plausible way to reach exactly this group.

A practical model for this already exists in how this project's participants were recruited: Seniornett's datacafé sessions, where volunteers help older adults with questions about digital devices in person (see section 3.3.1). A blended version of SeniorKI could work similarly. Participants would use the platform at their own pace, with the option to attend a

scheduled datacafé session specifically set aside for working through the platform's content with a volunteer present. This would keep the platform's core concept of self-paced learning, while adding a human fallback for the users the design alone did not reach.

This addition is also in line with the curriculum's authors concept: Kaur et al. (2025) outlined a blended learning approach, with a combination of in-person sessions and online resources. This project's findings, particularly the action paralysis pattern, gives a specific, evidence based reason to restore this part of the curriculums original design.

### 5.3.3 Social sharing of results

Adding some form of social sharing was considered during development, such as a share button on the badge earned page, letting users share their achievements with others. This was not prioritized above other functionality, as existing gamification literature has found that the ability to be connected with other users was rated as only somewhat motivating for older users, compared to other drivers, such as seeing progress and achievements (Gellner and Buchem, 2022). This suggests that social features may not be the most effective use of limited development time for this specific target group. Even so, this remains a reasonable feature to consider in future development, and testing it with SeniorKI's own users would give a more specific answer than a general literature comparison can provide.

## 6 CONCLUSION

This thesis aimed to design, develop, and evaluate an accessible AI literacy learning platform, based on the curriculum proposed by Kaur et al. (2025), with gamification elements, tailored to Norwegian adults aged 60 and above. The result is SeniorKI, a publicly available web platform implementing a five-module curriculum, evaluated through two rounds of formative usability testing and a summative evaluation with 7 completers.

The main research question asked to what extent the platform supports older adults' practical AI understanding, how usable participants find it, and to what extent its gamification elements encourage engagement and continued use. The answer is mixed across these three parts. On understanding, five of seven completers improved their knowledge test score, and all seven reached the maximum possible score on the post test. A Wilcoxon signed-rank test on the five participants with a non-zero change showed a statistically significant improvement using a one-sided test ( $p = .031$ ), though this significance did not hold under a two-sided test (.063) or when one participant's result was excluded due to a plausible, if unlikely and unconfirmed, information transfer concern (see section 5.1.3). The direction of the result is consistent and encouraging, but the small sample means it should be read as a promising signal rather than a confirmed effect.

On usability, the platform performed strongly. The average System Usability Scale score was 89.3, in the "Best imaginable" range on standard benchmarks, with participants rating the platform's learnability particularly highly. Qualitative feedback pointed to clear instructions and an appropriate level of difficulty as specific strengths. Two rounds of usability testing also identified and addressed several concrete usability barriers before the platform reached its final form. This included a per-module progress bar that functioned as unnoticed visual clutter, and topic pages that initially allowed users to advance past content they had not yet read. Not every usability issue was resolved, however: a pattern of hesitation bordering on giving up, observed independently in the oldest, least digitally experienced participants in both formative rounds, was not addressed by any design change identified during this project (see section 5.1.2).



On gamification and engagement, the evaluation is less conclusive. No summative participant used the platform across more than two days, and three completed the entire process, including the full course content and the post-test, in under an hour. This means the evaluation measured a single, short first encounter with the platform for many participants, rather than the sustained use over the two-week period the evaluation was intended to measure. As a result, this thesis cannot confidently say whether the platform's stars, badges, and progress overview succeeds at encouraging continued use over time, which was their intended purpose. What can be said, is that participants who commented on these elements consistently valued clear feedback on quiz answers more than the reward elements themselves, which is a pattern also found in prior research on gamification for older adults (Gellner & Buchem, 2022).

Against the project's four objectives, three were substantially achieved. The Kaur et al. (2025) curriculum was implemented as a functional, interactive platform, though without the practical activities the curriculum suggests for each module. This was a deviation made due to time constraints rather than design choice (see section 4.4). The platform was designed using universal design principles and evidence based guidelines for older adults, evaluated through automated and manual accessibility checks, though without a dedicated screen reader evaluation. The platform was evaluated iteratively across two formative rounds and a summative evaluation, though the second round fell short of the recommended five-participant target. The fourth objective, exploring whether gamification encourages engagement and continued use, was investigated but left largely unanswered by the available data, for the reasons discussed above.

Several directions for future work follow directly from these findings (see section 5.3 for the full discussion). Implementing the practical, hands-on activities specified in the source curriculum but not built within this project's timeframe is likely the single most valuable next step, both to make the platform closer to its curriculum basis, and because creativity-based activities were found to be a strong motivator in existing research. A blended format, pairing the platform's self-paced structure with optional in-person support through an organization such as Seniornett, could help reach the users who struggled most in this project's testing. A social sharing feature, suggested during development but not implemented, remains a

reasonable addition worth testing with users rather than deciding on the basis of general literature alone.

Beyond its research contributions, this project also has practical value. SeniorKI is a working, publicly available platform, not a prototype or minimum viable product. It is, to my knowledge, the first working implementation of Kaur et al.'s (2025) curriculum. The accessibility work, the design changes made across two iterations of testing, and the restrained approach to gamification documented throughout this thesis may be of use to others developing similar tools for older adults, which is a target group whose needs remain underrepresented in both AI literacy education and gamification research.

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## APPENDICES

### 1. Appendix A: SIKT approval

Confirmation of ethics approval (reference number: 697349) granted by Sikt (the Norwegian Agency for Shared Services in Education and Research) prior to data collection for the summative evaluation.

See attached file/image: *SIKT\_approval.pdf*

### 2. Appendix B: Source code

SeniorKI source code repository, tagged at the version corresponding to this thesis.

Link: <https://github.com/alexandermora1/ai-literacy-for-older-adults>, commit f31b012, tag v1.0-thesis

### 3. Appendix C: Usability test protocol: Iteration 1

Task list and observation guide used during the first round of formative usability testing (Figma prototype), conducted with five participants.

See attached file: *Usability\_test\_protocol\_1.pdf*

### 4. Appendix D: Usability test protocol: Iteration 2

Task list and observation guide used during the second round of formative usability testing (coded prototype), conducted with three participants.

See attached file: *Usability\_test\_protocol\_2.pdf*

### 5. Appendix E: Pre-test questionnaire (Nettskjema)

Blank pre-test questionnaire used in the summative evaluation, including the knowledge test and project information/consent text.

See attached file: *Nettskjema\_pre.pdf*

### 6. Appendix F: Post-test questionnaire (Nettskjema)

Blank post-test questionnaire used in the summative evaluation, including the knowledge test, System Usability Scale, and open-ended feedback questions.

See attached file: *Nettskjema\_post.pdf*

### 7. Appendix G: Curriculum content: Original version

Written curriculum content for all five modules, prior to language adaptation for the target group.

See attached file: *Curriculum\_original.pdf*



## 8. Appendix H: Curriculum content: Adapted version

Written curriculum content for all five modules, following language adaptation for the target group.

See attached file: *Curriculum\_adapted.pdf*

## 9. Appendix I: Summative evaluation data

Anonymized participant data from the summative evaluation, pseudonymized as P1–P9, including demographic information, pre/post knowledge test scores, System Usability Scale scores, and open-ended feedback responses.

See attached file: *Summative\_evaluation\_data.xlsx*

## 10. Appendix J: Language review prompt

Prompt given to Claude (Anthropic) for the language-level review of curriculum content, describing the target group and intended tone, used as a fallback after the planned review by Seniornett course leaders did not take place due to time constraints.

«Du er en tekstforfatter som hjelper med å forbedre læringsinnhold for voksne over 60 år som er nybegynnere på kunstig intelligens. Forbedre teksten jeg gir deg etter disse reglene, men ikke endre teksten hvis det ikke er behov for det.

Språk og tone

- Skriv på norsk bokmål i andre person ("du", "deg", "din")
- Bruk en vennlig, rolig og respektfull tone, ikke nedlatende eller barnslig
- Målgruppen er eldre, så gå ut ifra at de har et generelt godt ordforråd, men ikke når det gjelder tekniske IT-begreper.
- Bruk alltid forkortelsen "KI", ikke "AI"
- Unngå faguttrykk der det er mulig. Hvis et faguttrykk må brukes, forklar det umiddelbart i samme eller neste setning med enkle ord

Struktur og lesbarhet

- Del teksten inn i korte avsnitt på maks 3–4 setninger
- Bruk aktiv form fremfor passiv ("KI hjelper deg" fremfor "du hjelpes av KI")

Innhold

- Ikke introduser nye begreper eller informasjon som ikke finnes i originalteksten
- Behold alle faktapåstander fra originalteksten. Ikke legg til eller fjern faglig innhold.

Returner den forbedrede teksten i et nytt word-dokument.»

